

AI for Agriculture: Methane Emissions Gap-Filling, Forecasting, and Scenario Projection

Nicholas Tobias

Supervisors:

Paul Harris, Rothamsted Research

Varun Varma, Rothamsted Research

Andrea Karlova, UCL Computer Science

Faculty of Engineering

Department of Computer Science

University College London

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Abstract

Methane (CH_4) has approximately 28 times the 100-year global warming potential of CO_2 . UK agriculture contributes almost half (49%) of national methane emissions. Eddy Covariance (EC) provides ecosystem-scale observations of methane exchange over managed grasslands. However, EC records are often incomplete and characterised by episodic emission peaks. Gap-filling can reconstruct continuous historical flux records, while forecasting can anticipate future emissions and inform agricultural management.

This dissertation develops a three-stage machine-learning pipeline for gap-filling, forecasting, and scenario projection using data from three catchments at the North Wyke Farm Platform. Statistical, tree-based, neural, and tabular pretrained foundation models were benchmarked using held-out and temporal validation, supported by uncertainty quantification and permutation importance. Future climate trajectories and emission scenarios were then combined with alternative farm management conditions to explore possible changes in methane flux.

Overall performance follows a consistent pattern of tabular pretrained models performing best, followed by tree-based models, and locally trained neural networks generally underperforming. Nevertheless, sudden methane peaks remained difficult to anticipate for all models. Among the tested scenarios, livestock density produced the largest change in predicted annual CH_4 flux, followed by grazing duration, whereas varying fertiliser application rates produced little to no change in the modelled response.

Overall, pretrained foundation models for tabular data performed best across the predictive tasks, while livestock-associated variables formed the strongest recurring influence at the livestock-dominated catchments. However, severe target missingness, emission spikes, and the absence of future ground truth limit the operational utility of these models.

Keywords— methane flux – eddy covariance – time-series forecasting – gap-filling

Declaration

I, Nicholas Tobias, declare that this thesis has been composed by myself and that the work has not been submitted for any other degree or professional qualification. I confirm that the work submitted is my own, except where work that formed part of jointly authored publications has been included and appropriately referenced. The report may be freely copied and distributed provided that the source is explicitly acknowledged.

Nicholas Tobias

Signature

27 August 2026

Date

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List of Abbreviations

ACF: Autocorrelation Function

AIC: Akaike Information Criterion

ANN: Artificial Neural Network

AR: Autoregressive

Bi-LSTM: Bidirectional Long Short-Term Memory

CMIP6: Coupled Model Intercomparison Project Phase 6

CQR: Conformalized Quantile Regression

CV: Cross-Validation

DL: Deep Learning

DLinear: Decomposition-Linear Model

DOY: Day Of Year

EC: Eddy Covariance

FCH₄: Methane Flux

FCO₂: Carbon Dioxide Flux

GCM: Global Climate Model

GPP: Gross Primary Productivity

LSTM: Long Short-Term Memory

LSU: Livestock Unit

LightGBM: Light Gradient Boosting Machine

MAE: Mean Absolute Error

MASE: Mean Absolute Scaled Error

MDS: Marginal Distribution Sampling

MET: Meteorological Station

MICE: Multivariate Imputation by Chained Equations

ML: Machine Learning

MPIW: Mean Prediction Interval Width

NWFP: North Wyke Farm Platform

OLS: Ordinary Least Squares

PACF: Partial Autocorrelation Function

PCA: Principal Component Analysis

PICP: Prediction Interval Coverage Probability

PPFD: Photosynthetic Photon Flux Density

QC: Quality Control

QRF: Quantile Regression Forest

RF: Random Forest

RFm: Random Forest (met-driven variant)

RMSE: Root Mean Squared Error

SAITS: Self-Attention-based Imputation for Time Series

SARIMAX: Seasonal AutoRegressive Integrated Moving Average with eXogenous regressors

SMS: Soil Moisture Station

SSITC: Steady-State and Integral Turbulence Characteristics

SSP: Shared Socioeconomic Pathway

STL: Seasonal-Trend decomposition using Loess

SVM: Support Vector Machine

TFT: Temporal Fusion Transformer

TICA: Time-lagged Independent Component Analysis

TabICLv2: Tabular In-Context Learning Version 2

TabPFN: Tabular Prior-Data Fitted Network

UMAP: Uniform Manifold Approximation and Projection

UQ: Uncertainty Quantification

VPD: Vapour Pressure Deficit

WAPE: Weighted Absolute Percentage Error

XGB/XGBoost: Extreme Gradient Boosting

1 | Introduction

1.1 Research Objective

Methane (CH_4) is a powerful greenhouse gas, with a global warming potential approximately 28 times that of CO_2 over a 100-year horizon [4]. Agriculture is a major anthropogenic source, with enteric fermentation accounting for a substantial proportion of the sector’s greenhouse gas emissions [5, 6]. However, emissions from managed grasslands vary through time as livestock activity interacts with soil, meteorological, and management conditions. Understanding this variability therefore requires prediction of when ecosystem-scale emissions occur and how they respond to changes in farm management.

Eddy Covariance (EC) measurements provide a means of observing this integrated ecosystem-scale signal rather than methane emitted by individual animals alone. The North Wyke Farm Platform (NWFP) offers a rare record of half-hourly EC CH_4 flux from managed beef-cattle grasslands, accompanied by meteorological, soil, hydrological, and livestock-management data [2, 7]. This creates an opportunity for data-driven methane prediction, but also a difficult modelling problem: the record contains extensive gaps, its inputs occur at different temporal resolutions, and its target is highly variable and dominated by episodic emission peaks.

Existing machine-learning research on EC CH_4 in a temperate grassland setting has primarily addressed gap-filling, in which missing observations are reconstructed within an observed period [8, 9]. Forecasting is a different task: predictions must be generated sequentially using only information available at the forecast origin. To the author’s knowledge, no previous study has benchmarked machine-learning forecasting of ecosystem-scale EC CH_4 flux at a managed temperate grassland. Moreover, forecasting alone cannot answer counterfactual questions about how emissions might change under alternative livestock-management or climate conditions. This “what-if” capability is also

an underdeveloped area of research in agricultural digital twins [10, 11].

The objective of this dissertation is therefore to determine how effectively machine-learning models can reconstruct, forecast, and project ecosystem-scale CH_4 flux at NWFP. It develops a three-stage pipeline for gap-filling the observed Eddy Covariance record, forecasting future flux, and projecting modelled responses under varying management and climate scenarios. Model benchmarking is accompanied by uncertainty quantification and interpretability so that predictive performance, confidence, and inferred emission drivers can be evaluated together.

1.2 Project Aims and Goals

Aims:

Advance research in machine learning for CH_4 flux prediction at managed grassland, with a focus on gap-filling, forecasting, and scenario projection under contrasting livestock-management conditions.

Goals:

1. Generation of a pipeline for gap-filling, forecasting, and scenario projection of CH_4 flux at NWFP.
2. Benchmarking of multiple model families (from classical statistical baselines, tree ensembles, deep sequence models, to recent tabular foundation models) for gap-filling and forecasting.
3. Apply uncertainty quantification (quantile machine learning) and interpretability methods (permutation importance) across gap-filling, forecasting, and scenario projection.

4. Generation of future projection under differing livestock-management practices based on inference of previously generated model and CMIP6 climate projections.

1.3 Project Overview and Contributions

This dissertation contributes three benchmarked results to the field:

1. **A gap-filling benchmark for EC CH₄ flux at a managed grassland** (Chapter 5), comparing established methods, published baseline replications, and newer tabular foundation model families, with uncertainty quantification attached. The best configuration reaches an hourly-resolution R^2 of 0.429 - 0.686 across all three towers, against a published ceiling of $R^2 < 0.1$ at hourly resolution for the same ecosystem and data type [9].
2. **A forecasting benchmark for EC CH₄ flux at a managed grassland** (Chapter 6), the first, to the author's knowledge, applied to this ecosystem-data combination at a managed temperate grassland, comparing statistical baselines, tree ensembles, and tabular foundation models under identical conditions, with calibrated uncertainty quantification and interpretability through permutation importance. The best configuration beats seasonal-mean baseline by a wide margin (MASE=0.691), though a dedicated error evaluation shows that spike events remain a challenge for all models tested.
3. **Scenario projection under contrasting management and climate conditions for EC CH₄ flux** (Chapter 7): the forecasting architecture developed in this dissertation is extended to produce varying CH₄ projections under livestock and management practices for 2 distinct climate scenarios (SSP2-4.5 and SSP5-8.5), based on CMIP6 climate projections.

1.4 Report Overview

The subsequent chapters are structured as follows:

- Chapter 2 ([Background and Related Work](#)) reviews the literature establishing the gap that this dissertation aims to address, as well as the North Wyke Farm Platform (NWFP).
- Chapter 3 ([Data](#)) describes the NWFP dataset, exploratory data analysis, and processing required.
- Chapter 4 ([Models and Methods](#)) outlines the methodology used for the project, including the models, evaluation metrics, cross-validation procedure, uncertainty quantification and interpretability.
- Chapter 5 ([Gap-Filling](#)) presents the results of the gap-filling benchmark;
- Chapter 6 ([Forecasting](#)) presents the results of the forecasting benchmark;
- Chapter 7 ([Scenario Projection](#)) presents the results of scenario projection.
- Chapter 8 ([Discussion and Conclusion](#)) discusses the principal findings, their limitations and directions for future work before concluding the dissertation.

2 | Background and Related Work

2.1 North Wyke Farm Platform and Eddy Covariance Measurement

2.1.1 Experimental Design and Management Regimes

The North Wyke Farm Platform (NWFP), operated by Rothamsted Research near Okehampton in Devon, UK, is a farm-scale National Bioscience Research Infrastructure established in 2010 to investigate the productivity and environmental sustainability of temperate livestock and arable farming systems. The platform covers approximately 63 ha and comprises three outdoor experimental systems, known as the Red, Green, and Blue farmlets, each covering approximately 21 ha. Each farmlet contains five hydrologically isolated catchments, giving 15 catchments across the platform in total. Catchments may comprise either a single field or multiple separately managed fields. They are hydrologically isolated through the combined effects of local topography, slowly permeable subsoils, and a 9.2-km network of French drains that directs runoff towards instrumented flumes [1, 2].

The three farmlets represent contrasting agricultural management systems. The Red farmlet operates as an arable system with crop rotations including wheat, oats, and field beans. The Green farmlet represents a conventional permanent-pasture system, supporting seasonal cattle and sheep grazing alongside silage production and inorganic fertiliser application. The Blue farmlet combines high-sugar perennial ryegrass with white clover, supporting outdoor livestock production while using biological nitrogen fixation to reduce reliance on inorganic nitrogen fertiliser. Together, the farmlets allow arable, permanent-pasture, and grass-clover livestock systems to be studied under the same wider environmental setting [1, 12].

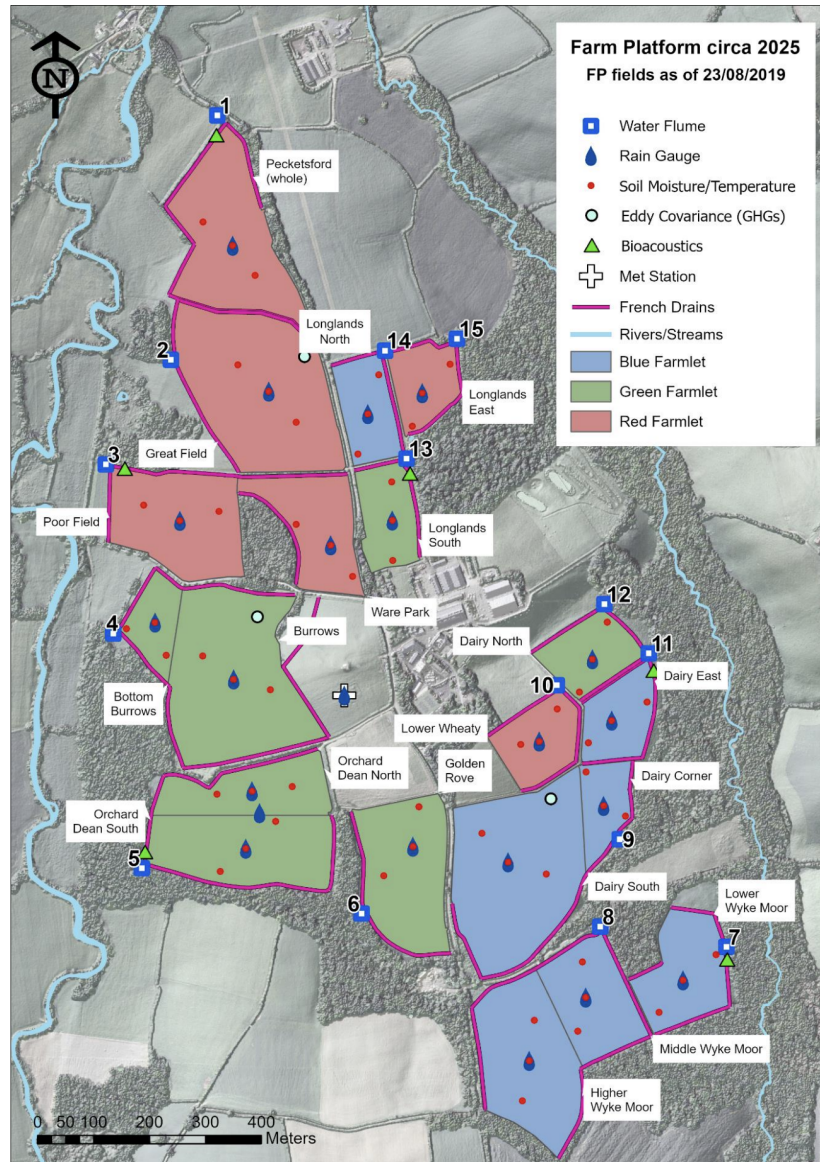


Figure 2.1: Layout of the North Wyke Farm Platform circa 2025, showing the Red, Green, and Blue farmlets; The EC systems considered in this dissertation are situated in Great Field within Catchment 2, Burrows within Catchment 4, and Dairy South within Catchment 9. Cited from [1]

This dissertation examines measurements associated with three selected catchments: Catchment 2 on the Red farmlet, Catchment 4 on the Green farmlet, and Catchment 9 on the Blue farmlet, shown in Figure 2.1. These locations are denoted T2, T4, and T9 throughout the dissertation. The notation identifies the catchment containing each EC site; it does not imply that NWFP comprises only three catchments. Moreover, the three locations differ substantially in land use, vegetation, livestock exposure, and measurement history. They should therefore be treated as distinct modelling contexts rather than spatial replicates of a single grassland treatment.

Table 2.1: Characteristics of the NWFP catchments associated with the Eddy Covariance Towers.

Site	Location	Land use and management	Modelling significance
T2	Catchment 2, Great Field, Red farmlet	Arable rotation including wheat, oats, and field beans. Great Field previously supported reseeded high-sugar perennial ryegrass, livestock grazing, and silage production.	Represents a pronounced grassland-to-arable transition rather than a continuously grazed pasture system.
T4	Catchment 4, Burrows, Green farmlet	Long-term permanent pasture managed through seasonal cattle and sheep grazing, silage production, inorganic fertiliser application, and the return of farmyard manure.	Represents the permanent-pasture system. Livestock presence within the EC footprint varies as animals rotate among fields.
T9	Catchment 9, Dairy South, Blue farmlet	High-sugar perennial ryegrass and white-clover pasture supporting cattle and sheep grazing and silage production, with biological nitrogen fixation reducing reliance on inorganic nitrogen fertiliser.	Represents the grass-clover livestock system, with temporally variable livestock presence and management activity.

Together, T2, T4, and T9 represent three distinct combinations of land use, vegetation, livestock exposure, and measurement availability. T2 captures a grassland-to-arable transition, T4 represents long-term permanent pasture, and T9 represents an established

grass–clover livestock system. These differences mean that observations from the three sites cannot be interpreted as spatial replicates of a common management regime.

2.1.2 Eddy Covariance Measurement and Flux Processing

Eddy Covariance (EC) estimates the vertical exchange of a gas between the land surface and atmosphere by combining rapid measurements of vertical wind velocity with corresponding changes in gas concentration. In simplified form, the turbulent flux of a gas c can be expressed as

$$F_c = \overline{w'c'}, \quad (2.1)$$

where w' is the deviation of vertical wind velocity from its averaging-period mean, c' is the corresponding deviation in gas concentration, and the overbar denotes averaging through time. EC therefore measures spatially integrated ecosystem exchange rather than emissions from an individual source.

Each of the three EC systems identified in Table 2.1 contains a Gill WindMaster Pro sonic anemometer and an enclosed LI-7200RS analyser measuring carbon dioxide and water vapour. Carbon-dioxide and water-vapour fluxes have been collected at all three systems since January 2017. Two open-path LI-7700 analysers were subsequently deployed for methane measurement from January 2018.

Because only two methane analysers were available, methane was not measured continuously at all three locations. The analysers were initially installed at T2 and T4 from January 2018 to June 2019. In July 2019, the analyser at T2 was relocated to T9 following the conversion of the Red farmlet to arable production. Consequently, T2 provides an earlier methane record from the Red pasture system, T4 provides the most temporally consistent record, and T9 provides a later record from the Blue grass–clover system [3].

These records are therefore neither continuous nor fully contemporaneous.

Raw EC measurements are processed using EddyPro software, which calculates gas fluxes and produces 30-minute averages recorded at the beginning and midpoint of each hour. Timestamps are maintained in Coordinated Universal Time using the analysers' GPS units. The resulting files follow the European Fluxes Database format and retain quality flags, friction-velocity measurements, and footprint information. Low-quality observations are not removed automatically and must therefore be identified during data preparation [3].

An EC observation represents a changing spatial footprint rather than a fixed average of the field or catchment containing the system. The location and extent of this footprint depend on wind direction, wind speed, atmospheric stability, and measurement height, and may occasionally extend beyond the field boundary. Recorded livestock presence at catchment level does not therefore guarantee that the animals were inside the EC footprint at a particular timestamp. The resulting FCH_4 measurement should be interpreted as integrated methane exchange within this changing footprint.

This measurement scale also distinguishes EC from systems such as GreenFeed. GreenFeed records methane released by an individual animal during discrete visits, whereas EC integrates methane exchange associated with livestock, manure, soil, vegetation, and other processes within its footprint [3, 13].

2.1.3 EC Methane Gap-Filling and the Forecasting Gap

EC records are rarely complete. Instrument maintenance, analyser servicing, quality-control filtering, insufficient atmospheric turbulence, and periods in which the estimated footprint falls outside the intended field can all remove observations. At NWFP, these operational gaps coexist with the longer structural absence created by relocating the methane analyser from T2 to T9. A continuous historical FCH_4 series must therefore be

constructed before flux summaries and downstream forecasting models can be developed. Zhu et al. provide the closest published same-site gap-filling comparison, having evaluated MDS and machine-learning approaches using EC methane measurements from the North Wyke managed pastures. All tested approaches produced an OLS-regression $R^2 < 0.10$ for FCH_4 : MDS achieved approximately $R^2 = 0.03\text{--}0.05$, while Random Forest performed relatively better but remained below $R^2 = 0.10$ [9]. These results establish the difficulty of gap-filling methane flux at NWFP and provide the principal published comparator for this dissertation.

Evidence from other ecosystems generally supports the use of tree-based methods. Irvin et al. found Random Forest and XGBoost to outperform neural and linear models across 17 FLUXNET-CH₄ wetland sites, although their uncertainty estimates were systematically too narrow [8]. Kim et al. similarly reported strong Random Forest performance and found lagged environmental variables particularly valuable for methane prediction [14]. Nevertheless, these studies are dominated by wetlands and rice paddies rather than managed livestock pasture. This motivates the same-site comparison of established gap-filling methods, newer model families, and agricultural management predictors undertaken in Chapter 5.

Importantly, these studies evaluate gap-filling rather than forecasting. Gap-filling reconstructs missing observations within a historical record and may use information from both before and after a gap. Forecasting instead predicts beyond a defined forecast origin using only information available at that point, with earlier predictions potentially becoming inputs during recursive rollout. The published gap-filling results therefore establish a relevant methodological baseline but do not demonstrate prospective predictive skill. To the author’s knowledge, no previous study has benchmarked multi-step EC methane forecasting at a managed temperate grassland; this forms the second modelling gap addressed by the dissertation.

2.2 Advances in Tabular Machine Learning

Following temporal alignment and feature construction, the modelling tasks in this dissertation take the form of supervised prediction over structured tabular data. The candidate model families nevertheless differ substantially: statistical models impose explicit temporal structure, tree ensembles learn non-linear partitions from the study data, conventional deep-learning models fit neural representations to the observed sequences, and tabular foundation models transfer a prediction procedure learned through large-scale pretraining. Two related questions therefore motivate the model roster used in this dissertation: whether recent tabular foundation models transfer effectively to EC methane data, and whether deep-learning methods outperform established statistical and tree-based alternatives in this setting.

2.2.1 Tabular Foundation Models

Tabular foundation models are pretrained across many synthetic prediction tasks and subsequently make predictions using labelled examples supplied as context. The original TabPFN applied this approach to small classification problems, while TabPFN v2 extended it to regression and larger tables [15, 16]. TabPFN is not inherently a time-series model, but Hoo et al. demonstrate that forecasting can be treated as tabular regression by representing temporal position, seasonality, lags, and other historical information as input features [17]. This formulation is directly relevant to the present dissertation, where EC methane forecasting similarly uses calendar, environmental, management, and autoregressive predictors.

TabICL was developed to extend in-context tabular learning to larger datasets [18]. TabICLv2 subsequently introduced a redesigned synthetic-data generator, more scalable attention, and revised pretraining procedures, while supporting both regression and classification [19]. Synthetic pretraining may be valuable at NWFP because valid methane

observations are limited, but general tabular benchmarks do not establish performance under extensive missingness, non-stationarity, episodic emission peaks, or recursive temporal extrapolation. The suitability of TabPFN and TabICLv2 for EC methane modelling must therefore be determined through task-specific evaluation.

2.2.2 Deep Learning versus Traditional Machine Learning

Large tabular benchmarks do not establish universal superiority of deep learning. Grinsztajn et al. compared tree ensembles and neural architectures across 45 medium-sized datasets and found tree-based models to remain state-of-the-art, attributing this partly to their robustness to uninformative features and irregular response functions [20]. McElfresh et al. reached a more conditional conclusion from 19 algorithms and 176 datasets: neural and boosted-tree performance was often similar, modest boosted-tree tuning could matter more than model-family choice, and relative performance depended strongly on dataset characteristics [21]. This pattern is consistent with EC methane studies, in which Random Forest and XGBoost have generally provided stronger baselines than conventional neural alternatives [8, 14, 9]. These findings justify retaining tree ensembles as serious comparators without assuming that they must remain superior when richer predictors or pretrained models are introduced.

A comparable debate exists in time-series forecasting. Zeng et al. found that simple one-layer linear models outperformed several Transformer architectures across nine long-term forecasting datasets [22]. This does not imply that complex architectures are universally ineffective, but demonstrates that architectural sophistication cannot itself be treated as evidence of predictive skill. Tabular foundation models further complicate the comparison: TabPFN and TabICLv2 are neural models, but their capabilities are largely acquired through pretraining rather than learned exclusively from the study dataset. The relevant empirical comparison is therefore among statistical baselines, tree ensembles, locally trained deep-learning models, and pretrained tabular foundation models under identical

evaluation conditions.

2.3 Synthesis and Gap Statement

Existing research demonstrates that machine learning can reconstruct missing EC methane observations, but the closest North Wyke comparison reports weak predictive performance, and the wider evidence is dominated by wetlands and rice paddies, not grassland farming. Moreover, these studies evaluate gap-filling within historical records rather than multi-step forecasting beyond a defined forecast origin. To the author’s knowledge, no previous study has benchmarked methane flux forecasting at a managed temperate grassland.

Recent TabPFN and TabICL models have also not been evaluated for this combination of ecosystem, measurement type, and prediction tasks. It therefore remains unknown whether synthetic pretraining provides an advantage over established statistical, tree-based, and conventional deep-learning approaches under severe target missingness and temporal extrapolation. This dissertation addresses these gaps by comparing the model families under common, leakage-safe gap-filling and forecasting protocols, accompanied by uncertainty quantification and interpretability. The resulting forecasting architecture is then applied to bounded scenario projection under contrasting management and climate conditions.

3.1 Dataset Overview

The analytical dataset draws on five NWFP data sources: the site-wide meteorological station, catchment soil moisture stations, eddy-covariance greenhouse-gas measurements, livestock-location records, and field-event records. The greenhouse-gas source used here is the tower-based EC dataset. A separate GreenFeed dataset measures emissions from individual animals but is excluded because it does not represent the same tower-scale ecosystem exchange measured by EC. Table 3.1 summarises these sources, while detailed dictionaries describing downloaded raw variables and project-derived variables are provided in Appendix A.

Table 3.1: NWFP datasets used to construct the analytical dataset. Dataset scope follows the NWFP Data Guide and platform design guide [2, 1]

NWFP dataset	Availability and resolution	Contents
MET Station [23]	31/10/2011–2 Mar 2026; 15 min	Site-level wind speed, air temperature, wind direction, relative humidity, solar radiation and precipitation
Soil Moisture Stations (SMS) [24]	31/10/2011–4 Mar 2024; 15 min	Catchment-level precipitation, soil moisture at 10 cm and soil temperature at 15 cm
Eddy Covariance Greenhouse Gas [3]	1/1/2017–1/1/2024; 30 min	Tower-level FCH ₄ , FCO ₂ , SSITC (quality) flags and associated EC biometeorological and turbulence variables, including VPD, PPFD, RN, USTAR and SHF
Livestock [2]	21/3/2011–21/4/2026; daily	Catchment-level daily location counts for livestock and weight
Field Events [12]	5/1/2011–13/11/2025; event based, sparse	Field-level farm operations, grazing practices, silage, fertiliser additions

Following time alignment and hourly aggregation, the consolidated table contained 70,153 hourly timestamps and 449 measurement variables. The hourly gap-filling frame later uses 30 predictors per tower, while forecasting uses the separate daily predictor frame described in Chapter 6. EC target analysis is restricted to the period from 1 January

2017 to 1 January 2024.

3.2 Exploratory Data Analysis and Data Preparation

3.2.1 Methane Target Availability

Target availability was assessed over the common 61,344-hour EC analysis window from 1 January 2017 to 1 January 2024. Raw missingness indicates whether an hourly FCH₄ value is present in the consolidated data. The modelling target is stricter: an observation is usable only when its SSITC quality class is 0 or 1 and its flux lies within the project plausibility range of $[-500, 3000]$ nmol m⁻² s⁻¹. Table 3.2 distinguishes these two stages, while Figure 3.1 shows both the temporal behaviour of the usable target and its annual completeness.

Table 3.2: Availability of hourly FCH₄ observations over the common 2017–2023 EC analysis window. QC-valid observations pass the SSITC and plausibility filters used to construct the modelling target.

Tower	Raw observed (h)	Raw missing (%)	QC-valid (h)	Unavailable after QC (%)
T2	8,099	86.8	4,890	92.0
T4	30,485	50.3	19,469	68.3
T9	17,509	71.5	11,235	81.7

Methane-target coverage is limited across all towers, with 92.0% of common-window hours unavailable at T2, compared with 68.3% at T4 and 81.7% at T9. Missingness is temporally structured: T2 observations are concentrated in 2017–2019, after which its methane analyser was relocated to T9 following the Red farmlet’s conversion to arable production [3]. T2 results should be interpreted cautiously due to its smaller, temporally narrower sample.

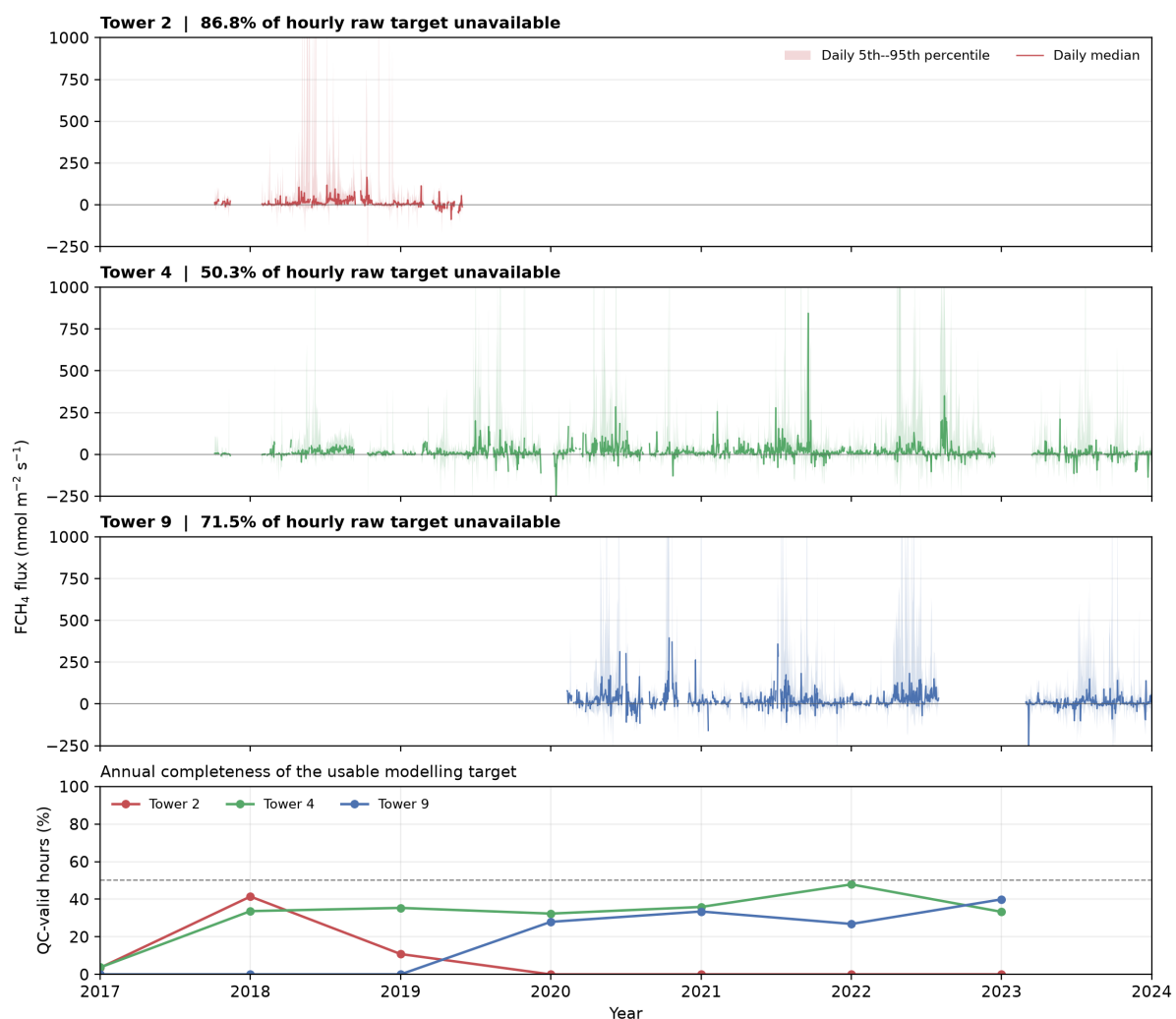


Figure 3.1: Daily median FCH₄ flux (line) and 5th–95th percentile range (shaded) at towers T2, T4, and T9, with hourly raw-data unavailability noted per panel. Bottom panel: annual completeness of the QC-valid modelling target. Note that y-axis has been limited to $[-250, 1000]$ nmol m⁻² s⁻¹ for readability.

3.2.2 Quality Control and Environmental Variable Gap-Filling

Quality screening was applied before reconstruction so that rejected observations could not act as anchors for filling. Hourly FCH_4 and FCO_2 observations were retained only when their SSITC class was 0 or 1; values outside $[-500, 3000]$ $\text{nmol CH}_4 \text{ m}^{-2} \text{ s}^{-1}$ and $[-100, 100]$ $\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$, respectively, were set to missing. The effect on the methane target is reported in Table 3.2. Before environmental-driver filling, friction velocity was restricted to $[0, 3]$ m s^{-1} and vapour-pressure deficit to $[0, 15]$ on the stored numerical scale because both fields contained extreme values. A separate low-turbulence flag was retained as a diagnostic rather than used to reject FCH_4 , since such filtering is not assumed to be a generally valid methane quality criterion.

Environmental-driver preparation combined two related procedures. The auxiliary-data calibration workflow, detailed in Algorithm 1 in Appendix A, was adapted from the North Wyke Farm Platform EC processing report [25]. The subsequent temporal filling hierarchy was implemented in Python and informed by the interpolation and mean-diurnal-course principles provided by `REddyProc` [26]; the R package itself was not invoked. The active data path used the central MET station for shortwave radiation, air temperature and wind speed, the corresponding catchment SMS for precipitation and soil conditions, and EC records for VPD, PPFD, net radiation, friction velocity and soil heat flux.

Following source selection and plausibility screening, internal gaps were linearly interpolated for up to 288 hours for air temperature, VPD, wind speed, soil moisture and soil temperature, and for up to two hours for the remaining drivers. Unresolved gaps were filled from the mean diurnal course at the same hour of day, first within a centred seven-day window and then, where necessary, within 14-, 28- and 60-day windows. Remaining gaps received the corresponding hour-of-day median and, only when that was unavailable, the series-wide median. Accepted observations were preserved throughout the temporal filling sequence. The resulting drivers are numerically complete while re-

taining local diurnal structure where nearby observations exist; values produced during longer sensor blackouts remain reconstructions rather than new measurements. This pre-processing was applied only to supporting model inputs; observed FCH_4 values were not imputed.

Table 3.3: Prior availability and synthetic-gap reconstruction accuracy of environmental drivers. Gap-CV R^2 is the mean across towers of the tower-level median over five gap-duration (1h, 4h, 32h, 288h, mixed) scenarios. Each scenario masks 25% of eligible observations across two repetitions for evaluation. The auxiliary-data calibration component is detailed in Algorithm 1 in Appendix A.

Driver	Available input (%)			After filling (%)	Gap-CV R^2
	T2	T4	T9	All towers	Mean across towers
Shortwave radiation	98.8	98.8	98.8	100.0	0.685
Air temperature	98.8	98.8	98.8	100.0	0.692
Vapour-pressure deficit	61.8	75.4	73.2	100.0	0.471
Photosynthetic photon flux	61.1	86.2	81.8	100.0	0.695
Net radiation	60.2	86.2	81.8	100.0	0.738
Wind speed	93.6	93.6	93.6	100.0	0.482
Friction velocity	81.2	88.2	84.4	100.0	0.000
Soil heat flux	43.3	86.2	81.8	100.0	0.532
Precipitation	83.1	85.7	62.5	100.0	0.012
Soil moisture	81.1	64.6	75.5	100.0	0.997
Soil temperature	83.7	65.6	78.9	100.0	0.993
FCO_2	50.5	64.2	58.5	100.0	0.745

Mean driver availability before reconstruction was 77.0% at T2, 84.5% at T4, and 82.8% at T9. The staged procedure achieved 100% numerical coverage across all 33 tower-driver series, but reconstruction accuracy varied considerably. Gap-CV performance was strongest for soil moisture and temperature, while precipitation and friction velocity showed limited skill. Thus, complete coverage does not imply equally reliable reconstruction across drivers. Their downstream contributions are evaluated in Chapter 5.

3.2.3 Temporal Analysis and Dimensionality Reduction

ACF and PACF were used to examine temporal dependence in the QC-valid observed FCH₄ series without reconstructing long target gaps. Weekly STL decomposition was also applied to the longest sufficiently complete daily segment at each tower. Full diagnostics are provided in Appendix Figures A.2 and A.3.

Pairwise associations with the target were calculated from the same 30-predictor frame used in the gap-filling experiments. Tower indicators were excluded because they are constant within each tower, and correlations were calculated only where observed FCH₄ passed the target quality and plausibility filters; gap-filled FCH₄ values were not used. Spearman’s rank correlation was selected because the target is highly skewed and spike-dominated, making a rank-based measure more representative of monotonic association than Pearson correlation. The calculation was performed separately for each tower using pairwise-complete observations. For a concise comparison, Figure 3.2 takes the five largest absolute correlations within each tower and displays their union, while retaining all three tower coefficients for every selected predictor. The complete 30-predictor matrix is provided in Appendix Figure A.1.

Figure 3.2 shows that livestock density has the strongest consistent positive association across the three towers ($\rho = 0.33, 0.20$, and 0.26 at T2, T4, and T9, respectively), with grazing status showing the same direction. Soil moisture and its lagged values are generally negatively associated with FCH₄, whereas soil temperature is positively associated, particularly at T2. However, all absolute correlations remain below 0.35 and their magnitudes differ substantially by tower. These coefficients therefore provide an initial screening of the feature space rather than a feature-importance result: they do not capture nonlinear interactions, and the closely related soil-lag variables share overlapping information.

Multivariate structure was examined using the same 11 environmental drivers employed

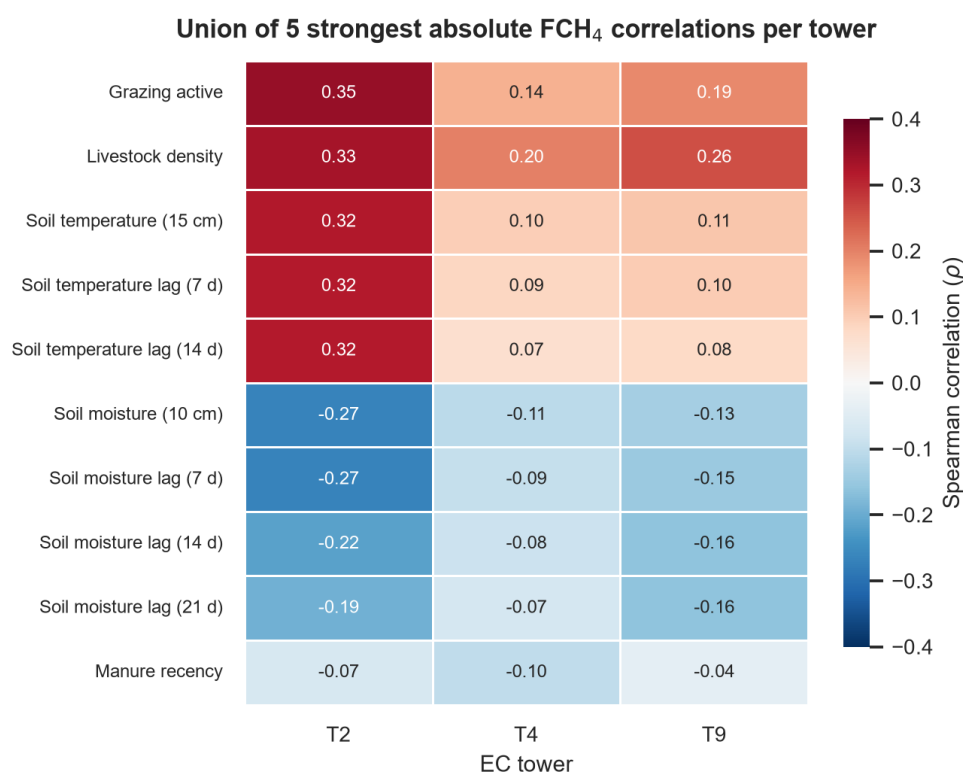


Figure 3.2: Strongest Spearman associations between observed QC-valid FCH₄ and the gap-filling predictors. The displayed variables are the union of the five largest absolute correlations selected independently within each EC tower; coefficients for all three towers are then shown for every selected variable. Positive values indicate that larger predictor values tend to accompany larger methane fluxes, whereas negative values indicate an inverse monotonic association.

by the gap-filling pipeline. TICA [27] was fitted at a 24-hour lag after standardisation, treating each tower as a separate contiguous trajectory so that lagged pairs could not cross tower or deployment boundaries. UMAP [28] was fitted to a tower-balanced sample of 3,000 hours using 30 neighbours and a minimum distance of 0.15. Neither analysis used FCH_4 , tower indicators, deterministic calendar encodings or precomputed lag features as inputs; methane availability was introduced only after fitting as a visual label.

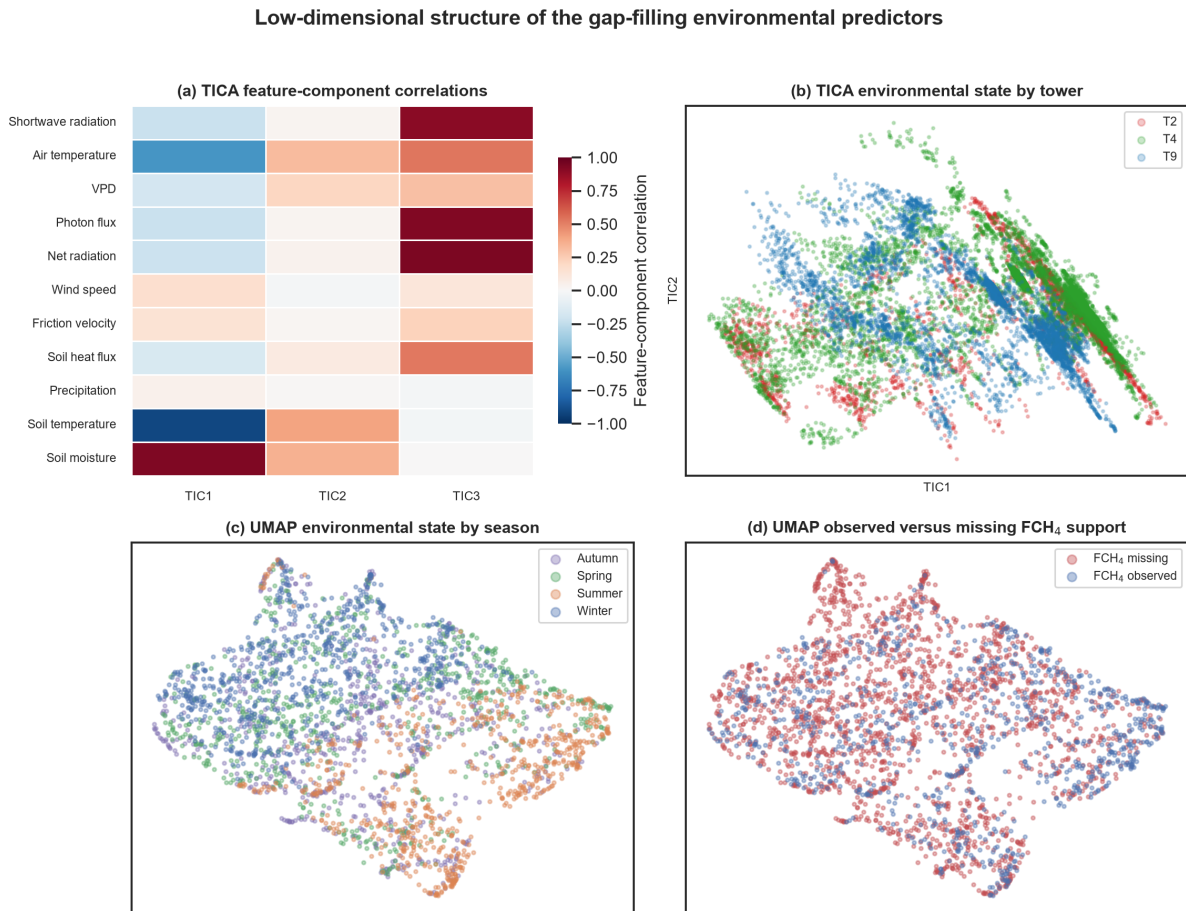


Figure 3.3: Low-dimensional structure of the environmental predictors used by the gap-filling pipeline. Panel (a) shows feature–component correlations for three TICA components at a 24-hour lag; panel (b) shows the first two TICA components by EC tower. Panels (c) and (d) show the same saved UMAP embedding coloured by season and by observed versus missing FCH_4 , respectively.

At a 24-hour lag, the first three TICA components had autocorrelations of 0.992, 0.970

and 0.869, implying timescales of 122, 33 and 7 days. The slowest component correlated strongly with soil moisture (0.93), soil temperature (0.91) and air temperature (0.59). UMAP showed moderate clustering by tower and season, but little separation between observed and missing FCH₄ hours. The two groups therefore occupy overlapping environmental space, though this does not establish that target missingness is random.

These diagnostics informed the final predictor design, but their low-dimensional representations were not retained. The observed temporal dependence supports the use of multi-scale lagged and rolling environmental variables, while fold-safe experiments found no consistent improvement from adding TICA components or UMAP-informed neighbour features. The final gap-filling and forecasting models therefore use the original environmental variables.

4 | Models and Methods

This chapter defines the common modelling and evaluation methodology used throughout the gap-filling, forecasting and scenario-projection experiments. Although these tasks use the same underlying NWFP data, they represent different prediction settings. Gap-filling reconstructs observations removed from within an historical record, forecasting predicts beyond a defined forecast origin, and scenario projection evaluates a fitted model under prescribed future environmental and management conditions. The corresponding validation procedures are therefore described separately before introducing the complete model roster, uncertainty-quantification framework and interpretability method.

4.1 Evaluation Methods and Metrics

The benchmark models optimise different internal objectives, including squared error, absolute error, likelihood-based objectives and quantile loss. Consequently, they were compared using common held-out evaluation metrics rather than their model-specific training losses. Table 4.1 summarises the principal distinction between the three modelling tasks.

4.1.1 Gap-Filling Cross-Validation

Gap-filling was evaluated as an interpolation problem over each tower’s usable measurement period. The evaluation domains were 1 October 2017–30 June 2019 for T2, 1 October 2017–31 December 2023 for T4, and 1 February 2020–31 December 2023 for T9. Only QC-valid observed FCH₄ values within these domains were eligible for masking.

Five synthetic-gap scenarios were used: very short gaps of one hour, short gaps of four hours, medium gaps of 32 hours, long gaps of 288 hours, and a mixed scenario in which gap length was sampled from these four durations. Non-overlapping calendar gaps were

Table 4.1: Summary of the evaluation settings used in the dissertation.

Task	Evaluation unit	Information available to the model	Primary metric
Gap-filling	Synthetic calendar gaps placed within the observed record	Remaining observations and supporting variables before and after each gap	R^2
Forecasting	365-day prediction from five historical forecast origins	Historical training data and realised supporting variables over the evaluation period	Seasonal-mean-scaled MASE
Scenario projection	Conditional prediction under prescribed climate and management trajectories	Historical training data and externally supplied scenario covariates	Relative change from the reference scenario

positioned randomly until approximately 25% of the available target observations had been masked. Two independently seeded repetitions were evaluated for each tower and gap scenario.

For every repetition, the masked FCH₄ observations were removed before fitting the model. Supporting environmental and management variables remained available, and the model could use historical context from either side of a gap. This reflects the intended use of gap-filling, in which later observations may be available when an earlier missing interval is reconstructed. It is therefore distinct from prospective forecasting and should not be interpreted as evidence of future predictive skill.

Metrics were first calculated independently for each repetition. The median over repetitions was then calculated within each gap-duration scenario, followed by the median across the five scenarios. This prevents a particularly large or easy synthetic gap from dominating the headline result. Where a combined value is reported across T2, T4 and T9, the tower-level results are averaged with equal weight so that the longer T4 record does not automatically dominate the comparison.

4.1.2 Forecasting Cross-Validation

Forecasting was evaluated through fixed-origin backtesting. Five forecast origins were defined on 16 December in 2018, 2019, 2020, 2021 and 2022. At each origin, a model was fitted using only data available on or before that date and generated a continuous 365-day daily prediction. Tower-origin combinations without sufficient historical or observed evaluation data were excluded rather than assigned artificial scores.

The resulting forecasts were evaluated against QC-valid observed FCH_4 , not against the reconstructed target series. The gap-filled historical series could be used to initialise autoregressive features, but predictions were scored only where a real observation was available. This prevents agreement with the gap-filling model from being mistaken for agreement with measured methane flux.

For recursive models, each predicted FCH_4 value was appended to the available history and used to construct the autoregressive inputs for subsequent days. No later observed methane value was inserted during the 365-day rollout. Direct tabular foundation-model configurations instead predicted from the available covariate representation without requiring the same recursive target-update mechanism.

Historical backtests supplied the realised supporting environmental variables over each future evaluation window. The resulting experiment is therefore a conditional methane-flux forecast: it evaluates the mapping from environmental and management conditions to FCH_4 , together with any recursive target dependence, rather than simultaneously evaluating a separate weather forecast. Future climate variables are supplied by the climate-scenario data in the scenario-projection experiment.

Performance was additionally divided into lead-time intervals of 1–7, 8–30, 31–90, 91–180, 181–270 and 271–365 days. This permits degradation with forecast horizon to be distinguished from overall performance. Aggregate metrics are weighted by the number of

observed target values contributing to each tower–origin–horizon cell, ensuring that cells containing very few real observations do not receive the same influence as well-observed cells.

4.1.3 Scenario-Projection Evaluation

Scenario projection has no observed future target and is therefore treated as a conditional projection rather than a forecast-validation exercise. The best-performing forecasting configuration is held fixed across all prescribed climate and management trajectories, which are compared with a common reference over the same projection period.

For scenario s , the relative change in mean predicted FCH₄ is calculated as

$$\Delta_s = 100 \left(\frac{\bar{\hat{y}}_s - \bar{\hat{y}}_{\text{ref}}}{\bar{\hat{y}}_{\text{ref}}} \right), \quad (4.1)$$

where $\bar{\hat{y}}_s$ and $\bar{\hat{y}}_{\text{ref}}$ denote the mean predictions under the scenario and reference trajectories, respectively. These values quantify modelled conditional differences and are not interpreted as validated forecasts of future methane flux.

4.1.4 Prediction Metrics

Let y_i denote an observed flux and $\hat{y}_i$ the corresponding model prediction for n valid evaluation observations. Mean absolute error and root mean squared error were calculated as

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (4.2)$$

and

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (4.3)$$

MAE represents the typical absolute error in the original flux units, whereas RMSE gives greater weight to the comparatively rare large errors associated with methane peaks.

Gap-filling primarily follows the OLS R^2 convention used by the published NWFP comparator [9]. In the implemented evaluation, this is the squared Pearson correlation between observations and predictions:

$$R^2(OLS) = \left[\frac{\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2}} \right]^2. \quad (4.4)$$

Unlike the residual coefficient of determination $1 - \sum (y_i - \hat{y}_i)^2 / \sum (y_i - \bar{y})^2$, Equation 4.4 is bounded between zero and one. It measures the strength of linear agreement but is comparatively insensitive to additive bias and scale error. MAE and RMSE are therefore retained as supporting metrics so that a model cannot be assessed from correlation alone.

Forecasting uses a seasonal-mean-scaled form of mean absolute scaled error. For each forecast origin, the reference prediction for a target date is the mean observed FCH₄ for the corresponding day of year within a circular ± 7 -day window, calculated exclusively from observations before the forecast origin. The reported metric is

$$\text{MASE}_{\text{seasonal}} = \frac{\text{MAE}(y, \hat{y})}{\text{MAE}(y, \hat{y}_{\text{seasonal}})}. \quad (4.5)$$

This applies the scale-free comparison principle of MASE [29] using the historical seasonal mean as the explicit reference forecast. Values below one indicate that the model improves on the seasonal mean, a value of one indicates equal error, and values above one indicate

inferior performance.

Weighted absolute percentage error was calculated as

$$\text{WAPE} = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{\sum_{i=1}^n |y_i|}. \quad (4.6)$$

WAPE is more stable than observation-wise percentage errors for this application because FCH_4 is signed and frequently lies near zero. Pearson correlation was also reported to describe agreement in temporal variation independently of absolute calibration. No single metric is treated as sufficient: MASE provides the primary forecasting comparison, while MAE, RMSE, WAPE and correlation identify differences in scale, peak sensitivity and temporal tracking.

4.2 Models Used

The benchmark intentionally includes simple baselines, conventional statistical models, tree ensembles, locally trained neural architectures and pretrained tabular foundation models. This breadth permits the claim that one model family outperforms another to be tested empirically under common evaluation conditions. Table 4.2 summarises how each model operates and where it is used. “GF” and “FC” denote gap-filling and forecasting respectively.

Table 4.2: Model families and their roles in the dissertation.

Model	Operating principle	Application
Training-set mean	Predicts the mean observed training target for every held-out row, providing a zero-information reference.	GF

Continued on next page

Table 4.2 – continued

Model	Operating principle	Application
MDS	Reconstructs flux from historical observations under similar meteorological conditions, followed by an expanding mean-diurnal-course fallback [30, 26, 9].	GF
MICE	Iteratively predicts each incomplete variable from the remaining variables. The implementation uses Bayesian ridge regression as the conditional estimator [31].	GF
HyperImpute	Extends chained-equation imputation by using an AutoML search to select a conditional estimator separately for each variable [32].	GF
Random Forest	Averages regression trees fitted to bootstrap samples with random feature selection at each split, capturing nonlinearities while reducing individual-tree variance [33, 8, 14, 9].	GF, FC
XGBoost	Builds regularised regression trees sequentially, with each tree correcting residual errors left by the existing ensemble [34, 8].	GF, FC
LightGBM	Uses histogram-based gradient boosting and leaf-wise tree growth to model nonlinear relationships efficiently [35].	GF, FC
ANN	Learns nonlinear feature combinations through successive fully connected layers trained by gradient-based optimisation [36, 14].	GF replication

Continued on next page

Table 4.2 – continued

Model	Operating principle	Application
SVM	Uses support-vector regression to fit a regularised function within an ϵ -insensitive error margin, with kernels permitting nonlinear relationships [37, 14].	GF replication
SAITS	Uses diagonally masked self-attention blocks and learned combination weights to reconstruct multivariate time-series gaps [38].	GF
LSTM and Bi-LSTM	Use gated recurrent memory to retain or discard temporal information. Bi-LSTM uses context on both sides of a historical gap, whereas forecasting LSTM operates causally during rollout [39, 40].	GF, FC
TFT	Combines recurrent processing, variable-selection networks, gating and temporal attention to represent local and longer-range forecasting relationships [41].	FC
DLinear	Decomposes a sequence into trend and residual components and forecasts each with a linear projection [22].	FC
SARIMAX	Combines autoregressive, differencing and moving-average terms with exogenous environmental and management regressors. Candidate orders are selected using AIC [42].	FC
TabPFN	Uses a transformer pretrained on synthetic tabular tasks to infer predictions from labelled context rows and unlabelled query rows [15, 16]. Temporal prediction is represented through engineered time-series features [17].	GF, FC

Continued on next page

Table 4.2 – continued

Model	Operating principle	Application
TabICLv2	Uses scalable tabular in-context learning with synthetic pretraining [18, 19].	GF, FC
Prediction ensemble	Averages predictions from complementary fitted models to reduce model-specific variance. Both tree–statistical and TabPFN combinations are evaluated [43].	FC

For locally fitted tabular models, missing predictor values were replaced using statistics estimated from the corresponding training data. Pooled models used tower indicators to share information while retaining tower-specific effects. Exact feature configurations, context lengths and hyperparameters are reported with the corresponding experiments; Table 4.2 defines only the common operating principles used throughout the results chapters.

4.3 Uncertainty Quantification

Point predictions alone do not distinguish confident predictions from periods in which multiple methane outcomes are plausible. Uncertainty was therefore represented through quantile-based machine learning and conformal calibration. For a conditional target distribution $Y \mid X = x$, the quantile model estimates

$$Q_\tau(x) = \inf \{y : P(Y \leq y \mid X = x) \geq \tau\}, \quad (4.7)$$

where $\tau \in (0, 1)$ denotes the desired quantile. The experiments use $\tau = 0.05, 0.50$ and 0.95 , producing a median point prediction and a nominal 90% interval.

Quantile models are trained or evaluated using pinball loss [44]:

$$L_\tau(y, \hat{q}_\tau) = \begin{cases} \tau(y - \hat{q}_\tau), & y \geq \hat{q}_\tau, \\ (1 - \tau)(\hat{q}_\tau - y), & y < \hat{q}_\tau. \end{cases} \quad (4.8)$$

The asymmetric weighting means that underprediction is penalised more heavily for a high quantile and overprediction more heavily for a low quantile. Minimising Equation 4.8 therefore estimates the corresponding conditional quantile rather than the conditional mean.

Several mechanisms were used to obtain initial prediction intervals. Quantile Regression Forest estimates conditional quantiles from the distribution of training outcomes represented within the fitted forest leaves [45]. XGBoost and LightGBM fit separate quantile-objective models for the 0.05, 0.50 and 0.95 levels, while SARIMAX obtains intervals from its fitted state-space forecast distribution. The TFT uses a quantile-output head, and TabPFN and TabICLv2 provide predictive quantiles from their pretrained regression distributions. For an ensemble, corresponding lower, median and upper predictions were combined using the same weights as the point predictions.

Raw model quantiles are not guaranteed to achieve their nominal coverage on NWFP data. Split-conformal calibration was therefore used to adjust the intervals using held-out residual information [46]. For symmetric split-conformal intervals, the calibration score for observation i is

$$s_i = |y_i - \hat{q}_{0.50}(x_i)|. \quad (4.9)$$

For a desired miscoverage rate $\alpha = 0.10$, the conformal margin is the finite-sample corrected $1 - \alpha$ quantile of the calibration scores. The calibrated interval is then

$$C_{0.90}(x) = [\hat{q}_{0.50}(x) - \hat{s}, \hat{q}_{0.50}(x) + \hat{s}]. \quad (4.10)$$

Forecasting calibration used a leave-one-anchor-out procedure. When one forecast origin was evaluated, residuals from the remaining four origins were pooled within the corresponding lead-time interval. The resulting conformal margin was then applied to the held-out origin. This preserves separation between calibration and evaluation while allowing interval width to vary between short and long forecast horizons.

Conformalized Quantile Regression was additionally used where the native quantile spread contained information that a symmetric interval around the median would discard [47]. Its nonconformity score is

$$s_i^{\text{CQR}} = \max \{ \hat{q}_{0.05}(x_i) - y_i, y_i - \hat{q}_{0.95}(x_i) \}, \quad (4.11)$$

giving the calibrated interval

$$C_{0.90}^{\text{CQR}}(x) = [\hat{q}_{0.05}(x) - \hat{s}_{\text{CQR}}, \hat{q}_{0.95}(x) + \hat{s}_{\text{CQR}}]. \quad (4.12)$$

For gap-filling, QRF and TabICLv2 intervals were evaluated on the synthetic held-out gaps. Calibration data were separated from the interval-evaluation data within each tower and gap-duration setting. For forecasting, calibration was separated by forecast origin and lead-time interval. Scenario intervals reuse margins learned from historical backtests; they quantify historically calibrated predictive dispersion but cannot guarantee coverage under an unobserved future distribution shift.

Let $\hat{l}_i$ and $\hat{u}_i$ denote the final lower and upper bounds of the interval evaluated for observation i . These may correspond to either the raw model quantiles or their conformally calibrated counterparts.

Three complementary interval metrics were reported. Prediction Interval Coverage Probability is

$$\text{PICP} = \frac{1}{n} \sum_{i=1}^n \mathbb{I} \left(\hat{l}_i \leq y_i \leq \hat{u}_i \right). \quad (4.13)$$

This measures the proportion of observations inside the interval. For a nominal 90% interval, values substantially below 0.90 indicate undercoverage and overconfidence, whereas values well above 0.90 indicate conservative intervals.

Mean Prediction Interval Width is

$$\text{MPIW} = \frac{1}{n} \sum_{i=1}^n \left(\hat{u}_i - \hat{l}_i \right). \quad (4.14)$$

MPIW measures sharpness. Narrower intervals are preferable only when their coverage remains comparable; an arbitrarily narrow interval with poor PICP is not useful. Mean pinball loss across the three quantiles measures the accuracy of the quantile predictions. PICP, MPIW and pinball loss are therefore interpreted jointly as coverage, sharpness and quantile accuracy, respectively.

4.4 Interpretability

Permutation-based prediction sensitivity was used as the common model-agnostic interpretability method. For each forecast origin and tower, the fitted model first produced an unmodified prediction vector $\hat{\mathbf{y}}$. Predictor j was then shuffled across the evaluation rows and the already fitted model was applied again without retraining.

Sensitivity was calculated as

$$I_j = \left| \frac{1}{n} \sum_{i=1}^n \hat{y}_i^{\pi(j)} - \frac{1}{n} \sum_{i=1}^n \hat{y}_i \right|, \quad (4.15)$$

where $\hat{y}_i^{\pi(j)}$ is the prediction obtained after permuting feature j . The score therefore measures the change in mean predicted FCH₄ caused by disrupting the information carried by that feature. Unlike conventional error-based permutation importance [33], it does not require an observed future target and can consequently be applied to conditional scenario projections.

Permutations were performed within each tower so that values were not mixed between T2, T4 and T9. Tower indicators were not permuted. Scores were calculated across the five forecast origins using deterministic seeds and were then summarised overall and by tower.

These scores measure model reliance rather than physical causality. Correlated predictors may share importance, while permutation may create combinations of variables that are uncommon in the observed data. Interpretation therefore focuses on stable feature groups and cross-tower patterns rather than small differences between individual rankings.

5.1 Overview

Chapter 3 showed that raw hourly FCH_4 was unavailable for 86.8%, 50.3% and 71.5% of the common analysis window at T2, T4 and T9, respectively. After applying the target quality and plausibility criteria, effective unavailability increased to 92.0%, 68.3% and 81.7% (Table 3.2). Chapter 3 reconstructed the supporting environmental variables to provide a complete predictor frame but deliberately left FCH_4 unfilled. Reconstructing this remaining target supports continuous retrospective analysis and downstream model configurations that require an uninterrupted methane history.

The experiments use the tower-specific hourly modelling frames constructed in Chapter 3. The response variable is QC-valid observed FCH_4 , while the expanded predictor frame combines current environmental conditions, carbon-cycle variables, calendar encodings, livestock information, environmental history and management events. Table 5.1 summarises these inputs.

The expanded 30-predictor frame was used for the principal tabular benchmark. Models sharing data across towers additionally received three tower-indicator variables, whereas TabICLv2 did not require them. MDS used only shortwave radiation, air temperature and VPD, while the meteorology-only Random Forest used the 11 current environmental variables. SAITS and Bi-LSTM omitted the eight explicit soil-lag variables because temporal context was supplied directly through their input sequences. Published-baseline replications retained their corresponding study-specific feature subsets.

All models were evaluated using the synthetic-gap protocol defined in Chapter 4. Consequently, the reported scores measure recovery of deliberately masked observations for which ground truth exists; they do not establish that predictions made during naturally

Table 5.1: Expanded predictor frame used in the gap-filling benchmark. Counts exclude the FCH₄ target and tower-indicator variables.

Feature group	Variables	Count
Current environmental conditions	SWIN, TA, VPD, PPFD, RN, WS, USTAR, SHF, precipitation, soil temperature and soil moisture	11
Carbon cycle	Gap-filled FCO ₂ , GPP and ecosystem respiration	3
Calendar	Cyclic hour-of-day and day-of-year sine and cosine encodings	4
Livestock	Livestock-unit density and grazing-presence indicator	2
Environmental history	Soil-moisture and soil-temperature lags at 168, 336, 504 and 672 hours	8
Management	Recency of cutting and manure application	2
Total		30

unobserved intervals are equivalent to measurements. The following sections compare the complete model roster against literature-aligned baselines and evaluate prediction-interval calibration.

5.2 Results

Unless otherwise stated, results are the median tower-specific R^2 values, calculated using the OLS convention defined in Chapter 4, under the common synthetic-gap protocol. Combined results are equal-weight averages over T2, T4 and T9, preventing the longer T4 record from dominating the comparison. Because this form of R^2 measures linear agreement rather than absolute calibration, the accompanying uncertainty analysis is used to qualify the headline ranking.

5.2.1 Full Model Comparison

MDS and the meteorology-only Random Forest provide literature-aligned reference points within the common benchmark, with Zhu et al. serving as the closest same-site comparator [9]. Adaptations of the broader Irvin and Kim benchmarks are retained in Appendix Table C.1 for context rather than treated as controlled cross-study comparisons.

Model or experiment	T2 R^2	T4 R^2	T9 R^2	Combined	Improvement over MDS
Training-set mean	—	—	—	—	—
MDS	0.072	0.038	0.054	0.055	0.000
RF (meteorology features only)	0.054	0.042	0.065	0.054	−0.001
MICE	0.104	0.128	0.110	0.114	+0.059
HyperImpute	0.567	0.352	0.362	0.427	+0.372
RF (additional features)	0.601	0.409	0.430	0.480	+0.425
LightGBM	0.583	0.418	0.425	0.475	+0.420
XGBoost	0.619	0.380	0.375	0.458	+0.403
SAITS	0.562	0.289	0.304	0.385	+0.330
Bidirectional LSTM	0.280	0.204	0.189	0.224	+0.169
TabICLv2	0.686	0.429	0.430	0.515	+0.460

Table 5.2: Full gap-filling model comparison under the common synthetic-gap protocol. Combined values are equal-weight averages of the three tower-specific R^2 scores; improvement over MDS is the difference between each model’s combined score and the combined MDS score. The training-set mean is constant, so its squared-correlation R^2 is undefined.

Table 5.2 shows that TabICLv2 produced the strongest overall linear agreement, leading at T2 and T4, matching the expanded Random Forest at T9, and attaining a combined R^2 of 0.515. The expanded Random Forest and LightGBM remained close at 0.480 and 0.475, respectively, whereas MICE and the two deep sequence models were weaker. The results therefore do not support a general claim that deep learning is superior to traditional machine learning: performance depended on the model architecture and its use of the available context. TabICLv2 is therefore retained as the gap-filling champion. Aggregate MAE and RMSE values for the same model comparison are reported in Appendix Table C.2.

Relative to MDS, the combined R^2 increased by 0.425 for the expanded Random Forest

and 0.460 for TabICLv2; their tower-specific gains ranged from 0.371 to 0.614. These are reported as additive changes rather than percentage gains because ratios against an R^2 value close to zero would exaggerate the apparent improvement.

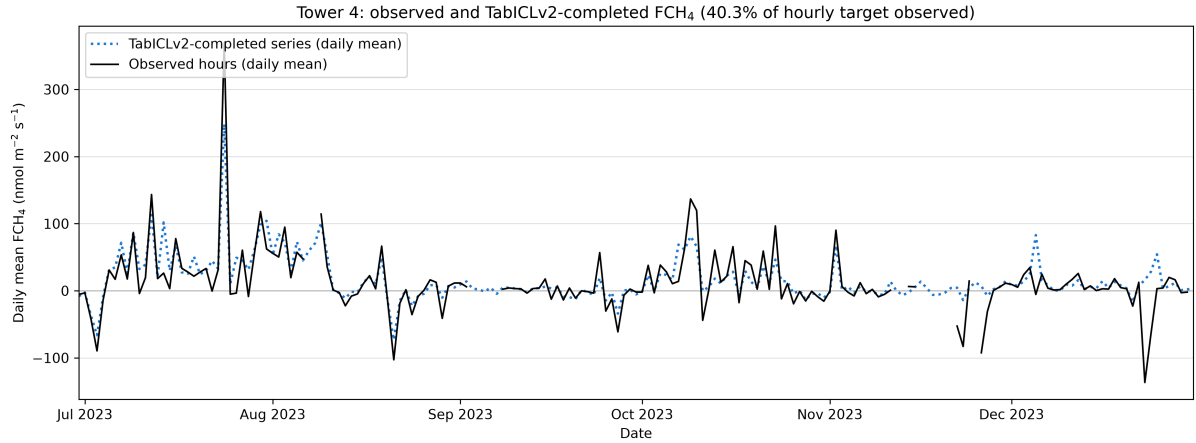


Figure 5.1: Daily means of QC-valid observed hours (solid black) and the TabICLv2-completed series (dotted blue) at T4 from July to December 2023, when 40.3% of the hourly FCH_4 target was observed. The completed series preserves observations and replaces missing hours with TabICLv2 estimates; the comparison is therefore a reconstruction diagnostic rather than held-out validation. Corresponding T2 and T9 reconstructions are provided in Appendix Figure C.1.

Figure 5.1 illustrates the operational effect of applying TabICLv2 to a sparse measurement period. The completed series provides a continuous daily trajectory and follows the broad observed variation, although several peaks are attenuated. Its overlap with the observed line is expected because measured hours are preserved rather than replaced. The figure therefore demonstrates the operational reconstruction while retaining the distinction between estimates and direct measurements.

5.2.2 Uncertainty Quantification

Raw 90% prediction intervals were obtained from Quantile Regression Forests for RFm and native quantile predictions for TabICLv2. Split-conformal calibration was then evaluated separately within each tower and gap-duration setting, following Chapter 4.

Model	Tower	n	Raw PICP	Calibrated PICP	Raw MPIW	Calibrated MPIW
QRF (additional features)	T2	4,961	0.938	0.893	163.8	147.6
	T4	19,485	0.917	0.901	196.1	186.8
	T9	11,279	0.902	0.903	214.2	216.2
	Average	35,725	0.919	0.899	191.4	183.5
TabICLv2 (quantile)	T2	4,961	0.918	0.904	134.5	130.1
	T4	19,485	0.899	0.899	181.6	181.2
	T9	11,279	0.888	0.893	214.5	217.2
	Average	35,725	0.902	0.899	176.8	176.2

Table 5.3: Raw and split-conformal 90% prediction-interval performance on the held-out calibration-test subset of the synthetic-gap experiment. PICP has a nominal target of 0.90; MPIW is reported in $\text{nmol m}^{-2} \text{s}^{-1}$. Tower rows are equal-weight means across the four gap-duration settings; averages weight all tower-gap cells equally, while n reports their total test observations.

Both raw interval methods were close to the 0.90 nominal target (Table 5.3). Calibration reduced average QRF coverage from 0.919 to 0.899 while narrowing MPIW from 191.4 to 183.5. TabICLv2 moved from 0.902 to 0.899 coverage while MPIW changed only from 176.8 to 176.2. Across the individual tower-gap cells, calibrated coverage ranged from 0.876 to 0.912 for QRF and from 0.888 to 0.910 for TabICLv2. Figure 5.2 deliberately shows the raw native interval from the production chain; the calibrated results in Table 5.3 come from the separate held-out synthetic-gap evaluation and were not applied to that chain.

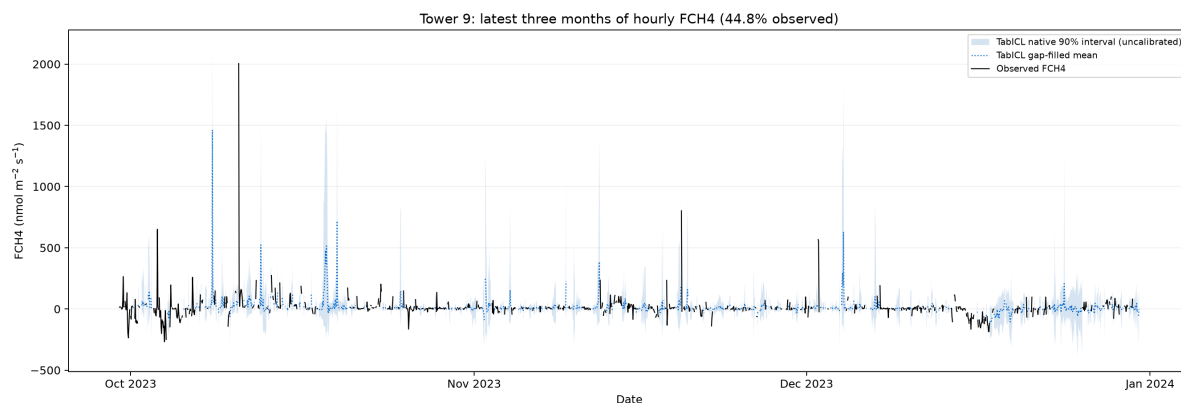


Figure 5.2: Three-month example of hourly FCH_4 reconstruction and native uncertainty at T9. The figure shows QC-valid observations, the TabICLv2 gap-filled mean and its uncalibrated native 90% prediction interval over the latest available three-month window, in which 44.8% of the target was observed. Corresponding T2 and T4 examples are provided in Appendix Figure C.2.

Overall, TabICLv2 provides the strongest point reconstruction and is retained as the gap-filling champion. Although calibrated coverage was close to the nominal target, the prediction intervals remained relatively wide, particularly at T9. Model identity and data-availability flags were therefore retained alongside the reconstructed FCH_4 series.

6.1 Overview

Forecasting extends beyond a fixed origin and therefore cannot use subsequent FCH_4 observations as context. The experiments follow the five 365-day fixed-origin backtests defined in Chapter 4. Supporting variables over each evaluation period are treated as known, so the results measure conditional methane-response forecasting rather than the additional error involved in forecasting weather or management. Performance is evaluated against QC-valid observed FCH_4 , not the reconstructed target series produced in Chapter 5.

The benchmark models use different predictor subsets. To avoid repeating a separate feature inventory for every model, Table 6.1 reports the complete predictor frame used by the TabPFN model. Its best configuration combines 52 daily environmental and management drivers with five tower and time indicators.

Best-performing forecasting model. The selected TabPFN and TabICLv2 models are equal-weight ensembles of three complementary variants. The first uses all pre-origin observations with the following probability-gated spike correction:

$$\hat{y}_t^{\text{corr}} = \hat{y}_t^{\text{base}} + 0.25 p_t^{\text{spike}} \left[\hat{y}_t^{\text{spike}} - \hat{y}_t^{\text{normal}} \right]_+, \quad [x]_+ = \max(x, 0).$$

Here, p_t^{spike} is the predicted probability of exceeding the tower-specific, pre-origin 95th percentile. The probability gates the correction, while the factor 0.25 retains only one quarter of the predicted positive spike excess to limit overcorrection. The other variants use the most recent 1,095 and 1,460 days, with the latter additionally applying tower-wise target normalisation. Equal-weight averaging reduces sensitivity to context length and spike-handling assumptions.

Predictor group	Included variables	Count
Meteorology and surface energy	Mean wind speed, friction velocity, mean/minimum/maximum air temperature, VPD, shortwave radiation, net radiation, PPFD, soil heat flux, precipitation and sine/cosine wind-direction components	13
Soil state and history	Current soil moisture and temperature; 7-, 14-, 21- and 28-day lags; and 7- and 14-day rolling means for both variables	14
Seasonality	Day-of-year sine and cosine encodings, growing-season indicator and winter indicator	4
Livestock and grazing	Livestock-unit, cattle, sheep, lamb and total-liveweight densities, grazing indicator and days since grazing	7
Hydrology and land use	Mean catchment flow, its 7-, 14-, 21- and 28-day lags and 7- and 14-day rolling means, plus the arable-field indicator	8
Management	Fertilizer-N application rate and recency, together with the recency of lime application, cultivation, cutting and manure application	6
Tower and absolute time	T2, T4 and T9 indicators, calendar year and days elapsed	5
Total	52 drivers and 5 tower/time indicators	57

Table 6.1: Complete daily predictor frame used by the TabPFN forecasting configuration. Simpler benchmark models use model-specific subsets of these variables.

6.2 Results

Model	Key Configuration	MASE (seasonal mean)	MAE	RMSE	WAPE	Correlation
TabPFN	Equal-weight blend of all-history spike-corrected, 1,095-day and 1,460-day TabPFN forecasts	0.6908	29.82	60.02	0.797	0.491
TabICLv2	Equal-weight blend of all-history spike-corrected, 1,095-day and 1,460-day TabICLv2 forecasts	0.7191	30.86	60.47	0.825	0.443
Ensemble	Equal-weighted mean of RF/XGBoost/LightGBM	0.8020	34.06	52.25	0.991	0.379
XGBoost	<code>max_depth=2</code> , <code>lr=0.02</code> , <code>n_estimators=400</code> , <code>min_child_weight=10</code>	0.8038	34.02	52.30	0.988	0.364
TFT	<code>d_model=32</code> , 4 attention heads, <code>dropout=0.1</code> , 28-day lookahead	0.8120	35.13	55.78	1.009	0.236
LightGBM	<code>num_leaves=7</code> , <code>min_child_samples=10</code> , <code>lr=0.02</code> , <code>n_estimators=400</code>	0.8166	34.81	52.83	1.014	0.371
Random Forest	<code>n_estimators=500</code> , <code>min_samples_leaf=10</code> , <code>max_features=0.5</code>	0.8411	35.17	52.91	1.045	0.380
SARIMAX	AIC-selected order ($p \in \{1, 2\}$, $d = 1$, $q \in \{0, 1\}$, per tower/anchor)	0.8741	36.11	54.35	1.082	0.348
LSTM	Hidden size 64, 28-day lookahead	0.9521	40.22	62.32	1.154	0.275
DLinear	Trend/seasonal decomposition, kernel=25, 28-day lookahead	1.1872	45.74	61.99	1.519	0.277
<i>Seasonal Mean (baseline)</i>		<i>1.000</i>	43.79	73.55	1.170	-0.004

Table 6.2: Forecasting performance against QC-valid observed FCH₄ across the evaluable 365-day fixed-origin backtests. MASE is scaled against the pre-origin seasonal-mean reference, for which MASE equals 1. Lower MASE, MAE, RMSE and WAPE indicate better performance, whereas higher correlation is preferable.

Table 6.2 shows that TabPFN achieved the strongest overall forecasting performance, recording the lowest MASE (0.691), MAE (29.82) and WAPE (0.797), together with the highest correlation (0.491). The conventional ensemble achieved the lowest RMSE (52.25). TabICLv2 ranked second by MASE at 0.719. The tree-based and ensemble models formed the next performance group, whereas LSTM and DLinear were the weakest configurations. DLinear was the only model with a MASE above 1, indicating worse absolute error than the seasonal-mean reference.

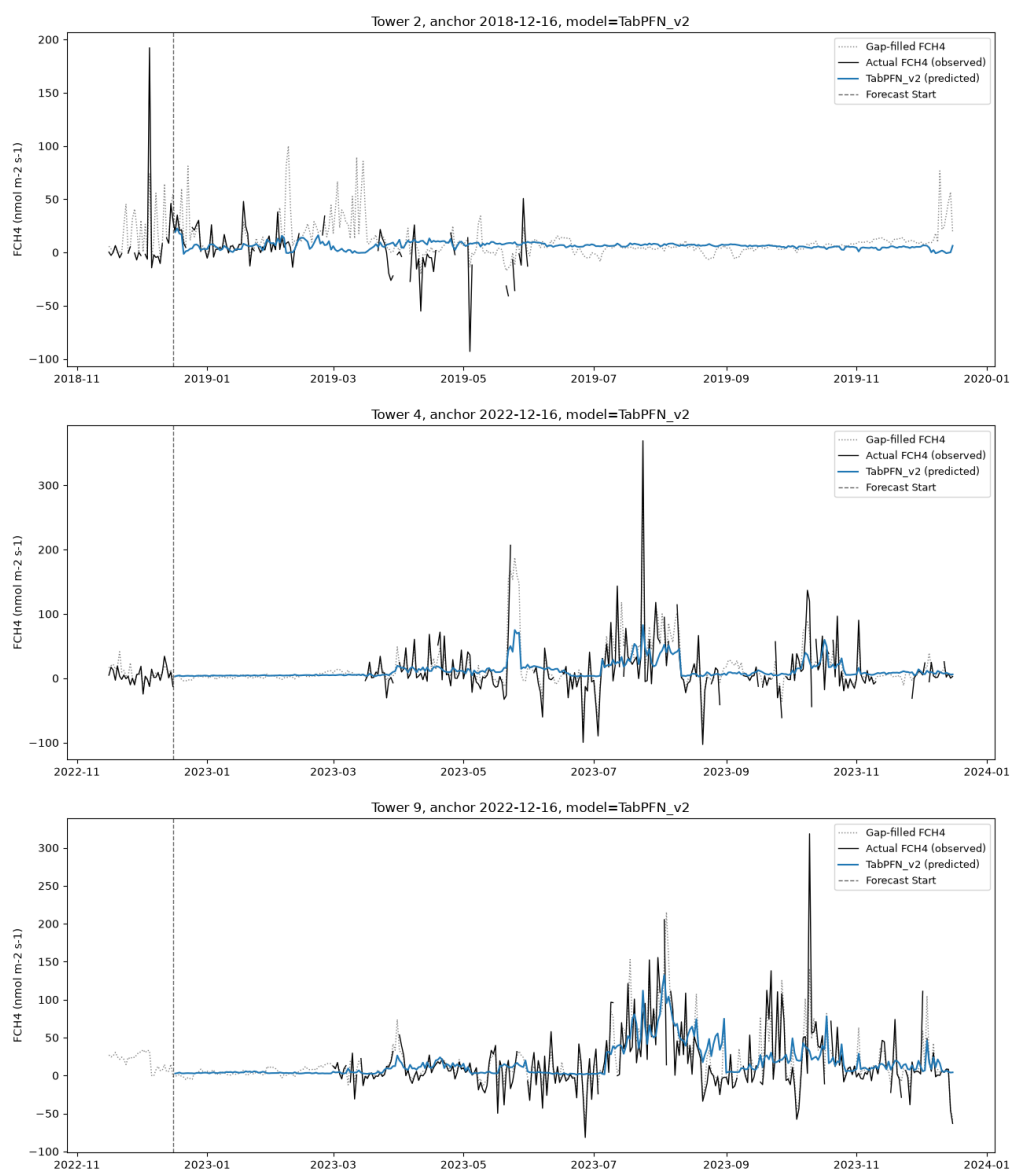


Figure 6.1: Representative 365-day conditional forecasts from the TabPFN model at T2, T4 and T9. The dashed vertical line marks the forecast origin; blue lines show predictions, solid black lines show observed FCH₄, and dotted grey lines show the gap-filled series for context.

6.2.1 Spike vs. Non-Spike Performance

The aggregate results conceal substantial differences between ordinary and high-flux periods. Table 6.3 separates the spike and non-spike observations for the TabPFN model using tower-specific thresholds estimated exclusively from observations preceding each forecast origin.

Metric	Non-spike ($n = 1,997$)	Spike ($n = 130$)
MASE	0.524	3.285
MAE	20.31	176.48
RMSE	30.01	210.12
Bias (prediction – observation)	+1.48	–176.48

Table 6.3: Forecasting performance of the TabPFN model during non-spike and spike periods. Spike thresholds are the tower-specific 95th percentiles estimated from pre-origin observed FCH_4 .

Forecast errors are concentrated in the spike periods. Non-spike observations achieved a MASE of 0.524 and MAE of 20.31, whereas spike observations reached a MASE of 3.285 and MAE of 176.48. Bias changed from only +1.48 during non-spike periods to –176.48 during spikes, showing that the dominant failure is severe underprediction of high-flux events. The temporal distribution of the evaluated spike observations is shown in Appendix Figure D.1.

6.2.2 Interpretability

Figure 6.2 shows the permutation-sensitivity ranking for the TabPFN model. Livestock-unit density, total-liveweight density and cattle density occupy the first three positions, followed by grazing status; the remaining environmental and management variables have substantially smaller effects. The tower-specific rankings in Appendix Figure D.2 show that this pattern is strongest at T4 and T9, whereas T2 relies primarily on environmental and management variables. Appendix Figure D.3 further separates the livestock related sensitivity by animal type.

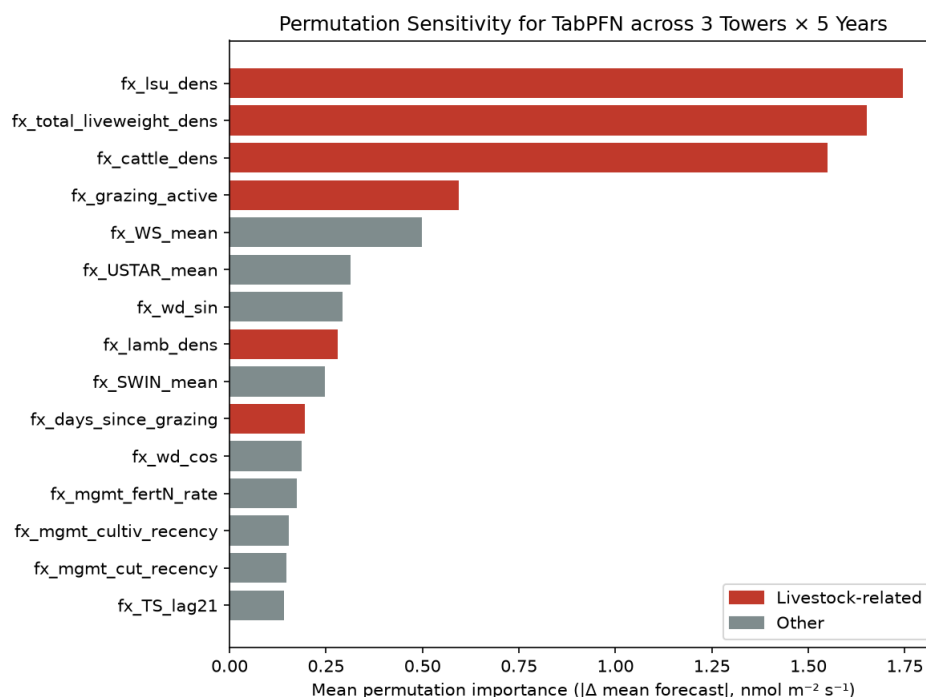


Figure 6.2: Permutation sensitivity of the TabPFN model, averaged across T2, T4 and T9 and five forecast origins. Each bar shows the absolute change in the mean forecast after shuffling one predictor within a tower; livestock-related variables are highlighted in red.

6.2.3 Uncertainty Quantification

Table 6.4 compares the raw and CQR-calibrated 90% prediction intervals for the TabPFN model. At T4 and T9, CQR reduced MPIW while shifting the mild raw overcoverage to modest undercoverage: PICP changed from 0.929 to 0.867 at T4 and from 0.928 to 0.871 at T9. T2 exhibited substantial raw undercoverage, but its sparse observed targets were insufficient to construct the leave-one-origin-out calibration set. Figure 6.3 illustrates the CQR interval defined in Chapter 4 for T4; the corresponding T9 result is provided in Appendix Figure D.5. Appendix Figure D.4 instead shows T2’s uncalibrated raw interval.

Tower	PICP		MPIW	
	Raw	CQR	Raw	CQR
T2	0.726	—	86.04	—
T4	0.929	0.867	187.13	178.14
T9	0.928	0.871	192.84	186.89

Table 6.4: Raw and CQR-calibrated 90% prediction-interval performance for the TabPFN model. PICP is the empirical coverage proportion and MPIW is reported in $\text{nmol m}^{-2} \text{s}^{-1}$. CQR values are unavailable for T2 because too few observed targets remained to construct its calibration set.

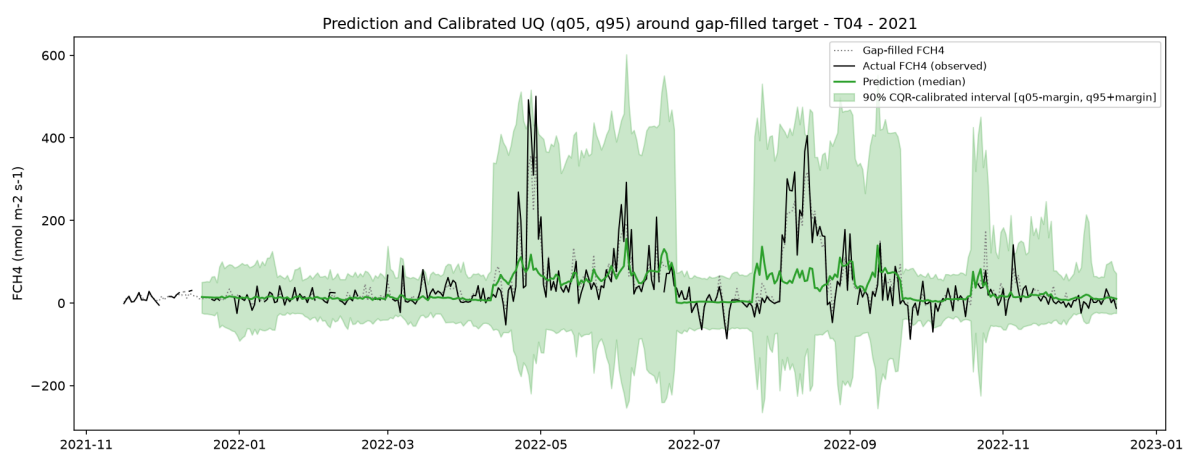


Figure 6.3: Representative 365-day TabPFN forecast at T4 from the 2021 forecast origin. The green line is the median forecast, the black line shows available observations, the dotted line provides the gap-filled series as context, and the shaded region is the 90% CQR-calibrated prediction interval across the complete forecast horizon. The corresponding T9 result is provided in Appendix Figure D.5.

Overall, TabPFN achieved the strongest average forecasting performance, but its errors remained concentrated during high-flux periods. Livestock-related variables dominated its sensitivity at T4 and T9, while CQR produced modest undercoverage and could not be applied at T2. These findings show that reliable long-horizon forecasting remains constrained by abrupt methane dynamics and sparse observations.

7 | Scenario Projection

7.1 CMIP6-Derived Climate Scenarios

Future weather conditions were obtained from the transient climate dataset developed by [48]. It provides daily simulations of minimum and maximum air temperature, precipitation and solar radiation for North Wyke from 2020 to 2090, generated using five CMIP6 Global Climate Models. This study selected ten realisations per GCM under SSP2-4.5 and SSP5-8.5, producing 100 climate trajectories, and restricted the projection horizon to 2050.

Unlike the historical NWFP data, these simulations provide site-level daily weather rather than tower- and catchment-specific observations. They contain neither FCH₄ nor the soil, hydrological, livestock and management measurements available historically. The scenario model was therefore restricted to predictors that could be obtained from the climate simulations or reconstructed consistently from the NWFP record. Variables unavailable from CMIP6 were reconstructed from historical NWFP seasonality. Livestock densities were represented using tower-specific day-of-year seasonal mean, while grazing and fertiliser variables were derived from historical presence and event patterns. These profiles were repeated across future years and modified according to each scenario.

The four simulated climate variables were bias-corrected against the local NWFP record using Tower 4 as the common reference. Per-GCM correction factors were estimated from the 2020–2030 SSP2-4.5 simulations: additive offsets were applied to minimum and maximum temperature, while multiplicative factors were applied to precipitation and solar radiation. The same corrections were applied under both pathways to preserve their relative differences. Mean temperature was subsequently derived from the corrected minimum and maximum values, and solar radiation was converted from MJ m⁻² day⁻¹ to W m⁻².

Table 7.1 reports the resulting correction factors. Raw simulations were cooler than the local record by 1.81–2.71°C for daily minimum temperature and 2.60–3.56°C for daily maximum temperature. Solar radiation required a modest upward adjustment of 3.4–8.3%, whereas precipitation was reduced to 22.5–24.0% of its raw value, indicating that the uncorrected simulations were approximately 4.2–4.4 times wetter than the NWFP reference.

GCM	$\Delta T_{\min}$ (°C)	$\Delta T_{\max}$ (°C)	SWIN ratio	Precipitation ratio
ACCESS-ESM1-5	1.81	2.60	1.034	0.225
CNRM-CM6-1	2.26	3.09	1.083	0.229
HadGEM3-GC31-LL	1.97	2.79	1.044	0.236
MPI-ESM1-2-LR	2.71	3.56	1.060	0.226
MRI-ESM2-0	2.29	3.03	1.043	0.240

Table 7.1: Per-GCM correction factors used to standardise the simulated climate variables against the Tower 4 NWFP reference. Temperature offsets were added to raw simulated values; solar-radiation and precipitation ratios were multiplied by raw simulated values.

7.2 Results

7.2.1 Less Informed Model Creation

This chapter’s scenario model uses this dissertation’s best forecasting model but restricted to the 13 features that CMIP6 climate scenario can actually supply (climate drivers, calendar encodings, livestock density, management practices) rather than the full 52-feature configuration used elsewhere in this dissertation (Chapter 6). This restriction carries an accuracy cost that has been quantified:

A restricted model is created using only the features available from the simulated climate dataset. This model has a 6.9% increase in MASE relative to the full-feature model (MASE 0.7383 vs. 0.6908), but remains sufficiently accurate for a projection task. This restricted model is used for the inference task in the sections that follow.

Configuration	Feature count	MASE (seasonal mean)
Full-feature model (Chapter 6)	52	0.6908
Restricted model	13	0.7383

Table 7.2: Accuracy cost of restricting the feature set to what CMIP6 climate scenario can supply. MASE scaled against the day-of-year seasonal mean baseline.

7.2.2 Livestock Density Sensitivity

Livestock density was varied along a four-level ladder: the historical baseline; half of the baseline density; a policy-informed reference level of 2.5 LSU ha⁻¹; and each tower’s own historical maximum. The reference level is drawn from UK Countryside Stewardship guidance for non-Severely Disadvantaged Area land [49].

Reported scenario responses are mean percentage changes in annual predicted FCH₄ relative to the matched baseline. They are averaged across the five GCMs and ten selected realisations per GCM.

SSP	Livestock level	T2	T4	T9
SSP2-4.5	Half (0.5×), all species	−27.5%	−14.9%	−26.9%
SSP2-4.5	Government reference (2.5 LSU/ha)	+4.6%	+0.9%	−2.6%
SSP2-4.5	Own historical maximum, all species	+101.2%	+138.3%	+184.3%
SSP5-8.5	Half (0.5×), all species	−27.2%	−15.0%	−26.9%
SSP5-8.5	Government reference (2.5 LSU/ha)	+4.6%	+0.9%	−2.6%
SSP5-8.5	Own historical maximum, all species	+99.7%	+138.1%	+181.9%

Table 7.3: Mean percentage change in annual predicted FCH₄ relative to the matched baseline livestock scenario, averaged across five GCMs and ten realisations per GCM.

Livestock density produced the largest modelled response in the scenario grid. Halving stocking density reduced predicted annual FCH₄ by 15–28%, while raising it to each tower’s historical maximum increased predicted flux by 100–184%. The policy-informed reference level of 2.5 LSU ha⁻¹ was close to the historical baseline at T2 and T4, but below the baseline at T9, producing only small changes. Responses at each livestock level were

similar under SSP2-4.5 and SSP5-8.5, indicating that the modelled sensitivity to livestock density was comparatively stable across the two climate pathways. Illustrative monthly trajectories are provided in Appendix Figures [E.1](#) and [E.2](#).

7.2.3 Grazing and Fertiliser Sensitivity

SSP	Intervention	T2	T4	T9
SSP2-4.5	Extend grazing season by 4 weeks	−3.8%	+19.9%	+20.0%
SSP2-4.5	Fertiliser frequency +50%	−8.4%	≈ 0.0%	≈ 0.0%
SSP2-4.5	Fertiliser rate +50%	−1.7%	−2.0%	≈ 0.0%
SSP5-8.5	Extend grazing season by 4 weeks	−3.7%	+19.9%	+19.9%
SSP5-8.5	Fertiliser frequency +50%	−8.2%	−0.1%	≈ 0.0%
SSP5-8.5	Fertiliser rate +50%	−1.7%	−2.0%	≈ 0.0%

Table 7.4: Mean percentage change in annual predicted FCH₄ relative to the matched historical scenario, averaged across five GCMs and ten realisations per GCM.

Extending the grazing season by four weeks increased predicted annual FCH₄ by approximately 20% at T4 and T9, whereas T2 produced a small negative response of approximately 4%. Fertiliser interventions produced smaller and less consistent responses: increasing application frequency reduced predicted flux by approximately 8% at T2 but had little effect at T4 and T9, while increasing the application rate changed predictions by no more than 2%. The similarity between SSP2-4.5 and SSP5-8.5 indicates that these modelled management sensitivities were stable across the two climate pathways. Illustrative trajectories are provided in Appendix Figures [E.3–E.6](#).

8.1 Main Findings

8.1.1 Overview of Model Performance

- **Gap-filling:** TabICLv2 is best (combined hourly $R^2 = 0.515$, Chapter 5), leading at T2/T4 and tying RF at T9. The two neural models nevertheless underperformed compared to the leading tree-based and foundation-model approaches: bidirectional LSTM ($R^2 = 0.224$) and SAITS ($R^2 = 0.385$).
- **Forecasting:** TabPFN is best (MASE= 0.6908, Chapter 6), while TabICLv2 ranks second (MASE= 0.7191). The weakest were again the locally trained deep-learning models: DLinear (MASE= 1.1872, the only model worse than the seasonal-mean baseline) and LSTM (MASE= 0.9521).
- **Tabular foundation models excel.** TabPFN and TabICLv2 are pretrained on millions of synthetic tabular datasets and predict via in-context learning, fitting no dataset-specific parameters [15, 16, 18, 19]. This prior-fitted-network approach is specifically effective on small, incomplete datasets [16] and may provide an advantage on NWFP’s severely incomplete record.
- **Deep learning does not perform best.** Locally trained neural architectures, including LSTM, Bi-LSTM, DLinear and TFT train from scratch on NWFP’s own incomplete record, giving a small effective training-sample size. Off-the-shelf recurrent architectures without dedicated missingness-handling tend only to match classical baselines (RF, SVM) on incomplete time series rather than exceed them [50]. This is consistent with deep learning’s broader failure to reliably beat tree-based methods on tabular data [20].

- **Uncertainty quantification also favours the tabular foundation models.**

In gap-filling, TabICLv2 matches QRF’s calibrated coverage (PICP= 0.899) with sharper intervals (MPIW 176.2 vs. 183.5, Chapter 5). In forecasting, TabPFN’s raw intervals already sit close to nominal at T4/T9 (PICP= 0.929/0.928) without extensive calibration – though not at Tower 2, where too few observations remain to calibrate at all (Chapter 6).

8.1.2 Difficulty in Methane Flux Gap-Filling and Forecasting

- **Gap-filling fit remains modest despite the improved pipeline.** Even with enriched features and a tabular foundation model, the combined hourly R^2 reaches only 0.515 (Chapter 5), a large improvement over MDS ($R^2 = 0.055$), but still a modest absolute fit.
- **The livestock-dominant catchments are hardest to reconstruct.** T4 and T9, where livestock density dominates the FCH_4 signal, score *below* the combined average ($R^2 = 0.429/0.430$), while T2, driven by a land-use transition, scores highest ($R^2 = 0.686$). The catchments most central to this dissertation’s finding are the ones it reconstructs least well.
- **Forecasting error is dominated by spikes.** Section 6.2.1 shows the champion’s strong pooled MASE (0.6908) is earned almost entirely on ordinary days (non-spike MASE= 0.524); on spike days it rises to 3.285, over six times worse and a clear loss against baseline. No feature in this project reliably predicts spike occurrence.

8.1.3 Scenario Sensitivity

Livestock density produced the largest modelled response among the tested scenarios, with historical-maximum stocking levels increasing predicted annual FCH_4 by 100–184%. Extending grazing increased predictions by approximately 20% at T4 and T9, whereas

fertiliser interventions produced smaller and less consistent responses. These sensitivities were similar under SSP2-4.5 and SSP5-8.5, although they represent conditional model behaviour rather than validated causal effects.

8.2 Limitations

- **Tower 2 is the weakest-conditioned data source at every pipeline stage.** Its QC-valid coverage is lowest of the three towers (92.0% unavailable vs. 68.3%/81.7% at T4/T9, Chapter 3); it cannot support conformal calibration in forecasting (Chapter 6) due to data limitations.
- **Forecasting and scenario projection rely substantially on gap-filled variables.** Both build on Chapter 5’s reconstructed FCH₄ series, most of which is model-estimated, not observed (68.3–92.0% of hours). Downstream results therefore partly reflect the gap-filling model’s own behaviour.
- **UQ and scenario projection were not extended to the full model roster.** Gap-filling UQ covers only its top two models (QRF, TabICLv2, Chapter 5); forecasting UQ covers only its champion (TabPFN, Chapter 6) – extending further was not attempted given the compute cost.
- **CMIP6 climate simulations do not match NWFP’s own historical climate.** Raw air temperature runs 1.81–2.71°C too cool and precipitation $\sim 4.2\text{--}4.4\times$ too wet before correction (Chapter 7); the applied correction reduces but does not eliminate this as a residual uncertainty source.
- **Scenario projection has no ground truth to validate against.** No real CH₄ measurement exists for conditions that have never occurred at NWFP, and this dissertation makes no comparison against any process-based (mechanistic) methane model; its results describe conditional model behaviour, not a validated forecast.

8.3 Future Work

- **Rerun the experiment once extended Tower 2 data is released.** Tower 2’s usable record is expected to grow soon: 2025 data has already been collected and is pending release. Rerunning once available would test whether Tower 2’s results reflect a resolvable data-scarcity limitation (Section 8.2) rather than a persistent structural difference.
- **Livestock proximity to each tower’s footprint as an additional gap-filling feature.** This dissertation’s livestock-density feature is catchment-level, but an EC footprint shifts with wind direction, speed, and stability (Chapter 2), so catchment-level presence does not guarantee animals were within it. Proximity-weighted, footprint-aware livestock location via GPS or remote sensing could sharpen the single strongest predictor identified throughout this dissertation.
- **Benchmark tabular models against other eddy-covariance datasets.** This dissertation’s finding that pretrained tabular foundation models outperform locally-trained alternatives under severe missingness (Section 8.1.1) was established at a single site. Testing the same roster against benchmarks such as FLUXNET-CH4 [8] would test whether this generalises. However, FLUXNET-CH4 predominantly covers wetlands and rice paddies rather than managed grasslands, so this would specifically test the pretrained-model and environmental-driver findings.

8.4 Concluding Remarks

Overall, this dissertation developed and evaluated an integrated framework for methane-flux gap-filling, conditional forecasting and exploratory scenario analysis at the North Wyke Farm Platform. Pretrained foundation models for tabular data provided the strongest overall predictive performance, while traditional tree-based methods generally

outperformed locally trained neural networks. Livestock-associated variables also formed the most consistent influence across the forecasting and scenario experiments. These findings remain constrained by substantial target missingness, poorly predicted emission peaks and the absence of future ground truth, and the scenario responses should not be interpreted as causal effects. Nevertheless, the resulting benchmark provides a reproducible foundation for further methane modelling at North Wyke and other managed grassland sites.

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A | Data

The downloaded raw-data directory contains 186 annual CSV files from 26 logical dataset families. The tables below document the EC variables and project-derived features used by the modelling pipeline. The annual files are not perfectly schema-uniform: the 2017 greenhouse-gas file has a malformed header containing appended unit fields, subsequent greenhouse-gas files place units in a separate first data row, and the environmental-measurement schema decreases from 718 to 694 columns from 2022 as phosphorus streams for several catchments disappear. Annual query windows also overlap at some boundary timestamps. Compilation therefore removes duplicated timestamps and normalises the time index before hourly aggregation. The greenhouse-gas record ends on 1 January 2024, so its substantive observations cover 2017–2023. Later records are available for environmental, livestock and management sources, but not for EC fluxes.

Table A.1: Complete base-variable dictionary for the raw eddy-covariance, biomet and turbulence files. Each listed base variable is repeated for Towers 2, 4 and 9. Definitions, units and native resolution follow the NWFP EC greenhouse-gas guide [3]; treatment and analytical role describe processing in this project. Tower identifiers and repeated suffixes are omitted for readability.

Variable	Meaning	Unit and resolution	Treatment and analytical role
CO2	Carbon-dioxide mole fraction measured by the gas analyser	mmol CO ₂ mol ⁻¹ ; 30 min	Available raw concentration; not retained as a direct model feature
H2O	Water-vapour mole fraction measured by the gas analyser	mmol H ₂ O mol ⁻¹ ; 30 min	Available raw concentration; not retained directly
CH4	Methane mole fraction measured by the gas analyser	nmol CH ₄ mol ⁻¹ ; 30 min	Distinct from ecosystem-scale methane flux; not retained directly
TAU	and Momentum flux and its steady-state/integral-turbulence class	kg m ⁻¹ s ⁻² and quality categorical; 30 min	Available turbulence output; not retained directly
TAU_SSITC_TEST			
H and H_SSITC_TEST	Sensible-heat flux and its quality class	W m ⁻² and categorical; 30 min	Available energy-balance output; not retained directly

Continued on next page

Table A.1 – continued

Variable	Meaning	Unit and resolution	Treatment and analytical role
LE	and Latent-heat flux and its quality	W m^{-2} and cate-	Available energy-balance output; not
LE_SSITC_TEST	class	gorical; 30 min	retained directly
FC (FCO ₂)	Net ecosystem carbon-dioxide ex- change	$\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$; 30 min	Carbon-cycle predictor and par- titioning input; values outside [−100, 100] are rejected
FC_SSITC_TEST	SSITC quality class for each carbon-dioxide-flux estimate	Categorical: 0 high, 1 medium, 2 low quality; 30 min	Classes 0 and 1 are retained before FCO ₂ reconstruction
FCH ₄ (FCH ₄)	Methane flux measured by eddy covariance	$\text{nmol CH}_4 \text{ m}^{-2} \text{ s}^{-1}$; 30 min	Prediction target; classes 0 and 1 are retained and fluxes outside [−500, 3000] are treated as missing
FCH ₄ _SSITC_TEST	SSITC quality class for each methane-flux estimate	Categorical: 0 high, 1 medium, 2 low quality; 30 min	Provider quality field used to screen the target
WD	Horizontal wind direction	degrees; 30 min	Converted to sine and cosine compo- nents when used in forecasting
WS/WS_MAX	Mean and maximum horizontal wind speed	m s^{-1} ; 30 min	Mean wind speed is retained; the maximum is available but not re- tained
U/V/W_SIGMA	Standard deviations of the three wind components	m s^{-1} ; 30 min	Available turbulence diagnostics; not retained directly
USTAR	Friction velocity representing at- mospheric turbulence	m s^{-1} ; 30 min	Predictor and diagnostic; values out- side [0, 3] are removed, while the de- rived low-turbulence flag is not ap- plied to FCH ₄
MO_LENGTH/ZL	Monin–Obukhov length and di- mensionless stability ratio z/L	m and dimension- less; 30 min	Available atmospheric-stability diag- nostics; not retained directly
FETCH_MAX, FETCH_70/80/90	Maximum fetch and distances con- taining 70, 80 and 90% of the mod- elled footprint contribution	m ; 30 min	Available footprint diagnostics; not retained as model features
PA	Atmospheric pressure	kPa; 30 min	Available meteorological variable; not retained directly
RH_0/RH_1	Relative humidity measured by the EC and biomet systems	%; 30 min	Available but excluded from retained feature sets in favour of VPD

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Table A.1 – continued

Variable	Meaning	Unit and resolution	Treatment and analytical role
TA_0 (TA)	Air temperature	°C; 30 min	Meteorological predictor and carbon-partitioning input; an external series is available
VPD	Vapour-pressure deficit, representing atmospheric moisture demand	hPa; 30 min	Retained predictor; a [0, 15] hPa plausibility bound is applied before gap-filling to remove implausible outlier readings
T_SONIC and sigma	Sonic temperature and its standard-deviation field	°C; 30 min	Available sonic-anemometer outputs; not retained directly
LWIN/LWOUT	Incoming and outgoing long-wave radiation	W m ⁻² ; 30 min	Available radiation components; not retained directly
PPFD	Incoming photosynthetic photon-flux density	μmol photons m ⁻² s ⁻¹ ; 30 min	Radiation and plant-activity predictor
RN	Net radiation at the land surface	W m ⁻² ; 30 min	Surface-energy predictor
SHF_1/2/3	Soil heat flux recorded by three plates	W m ⁻² ; 30 min	Sensor 1 is retained in the base feature set; all three are averaged in selected forecasting configurations
SWC_1/2/3	Volumetric soil-water content from three EC-biomet sensors	m ³ m ⁻³ ; 30 min	Available EC soil measurements; the adopted configuration instead uses catchment SMS observations
SWIN/SWOUT	Incoming and outgoing short-wave radiation	W m ⁻² ; 30 min	Incoming radiation is retained; outgoing radiation is not retained directly
TA_1	Secondary biomet air-temperature measurement	K; 30 min	Available raw measurement; not retained directly
TS_1/2/3	Soil temperature from three EC-biomet sensors	K; 30 min	Sensor 1 at Tower 9 supplies the cross-tower proxy only in the EC-only configuration; adopted catchment SMS temperatures are in °C

Table A.2: Dictionary of principal project-derived variables. These are not original NWFP portal columns: their definitions follow the repository’s preprocessing and feature-construction code, while the underlying raw measurements use the guide-based definitions in the preceding tables.

Derived variable	Meaning or derivation	Unit	Analytical role
Gap-filled FCO ₂	QC-valid observations are retained; missing values are reconstructed separately for each tower using a meteorology-based random forest	$\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$	Environmental predictor and basis for GPP and Reco
GPP	Gross primary productivity from Reichstein/Lloyd–Taylor partitioning; $\max(\text{Reco} - \text{NEE}, 0)$ and zero at night	$\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$	Plant-productivity predictor
Reco	Ecosystem respiration from the Lloyd–Taylor temperature response using global E_0 and seven-day-block R_{ref} refitting	$\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$	Ecosystem-respiration predictor
LSU density	$(1.0N_{\text{cattle}} + 0.1N_{\text{sheep}} + 0.05N_{\text{lamb}})/A$; the pipeline uses $A = 6.65$ ha for Catchment 2 and 7.75 ha for Catchments 4 and 9	LSU ha^{-1}	Standardised grazing-intensity predictor; the Catchment 2 constant represents its historical fenced area rather than its post-conversion cropped area
Grazing active	One when the summed livestock count is greater than zero; otherwise zero	Binary	Management-state predictor; the current implementation fills missing species counts with zero before calculating it
Current grazing-spell duration	Zero on non-grazing days; on grazing days, zero on the first day and incremented for each subsequent consecutive day	days	Run-length measure of the active grazing spell; despite its code name, it is not elapsed time since the last grazing event
Cutting recency	$\exp(-d_{\text{cut}}/21)$	Dimensionless [0, 1]	Retained management predictor
Manure recency	$\exp(-d_{\text{manure}}/30)$	Dimensionless [0, 1]	Retained management predictor
Hour sine and cosine	$\sin(2\pi h/24)$ and $\cos(2\pi h/24)$	Dimensionless	Diurnal timing predictors
Day-of-year sine and cosine	$\sin(2\pi d/365)$ and $\cos(2\pi d/365)$	Dimensionless	Seasonal timing predictors

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Table A.2 – continued

Derived variable	Meaning or derivation	Unit	Analytical role
Soil-moisture lags	Gap-filled soil moisture shifted by 168, 336, 504 and 672 hours	%	One- to four-week environmental-memory predictors
Soil-temperature lags	Gap-filled soil temperature shifted by 168, 336, 504 and 672 hours	°C for the adopted SMS configuration; K for the EC-only proxy	One- to four-week environmental-memory predictors
Three-sensor mean	SHF Arithmetic mean of SHF1, SHF2 and SHF3	W m ⁻²	Forecasting predictor in selected configurations
Wind-direction components	$\sin(\theta)$ and $\cos(\theta)$ for wind direction θ in radians	Dimensionless	Circular encoding that removes the discontinuity at 0°/360°
Tower indicators	One-hot variables <code>is_t2</code> , <code>is_t4</code> and <code>is_t9</code>	Binary	Preserve tower identity in partially pooled models

The auxiliary-data calibration procedure referenced in Section 3.2.2 is reproduced in Algorithm 1.

Algorithm 1 Auxiliary-data gap filling of meteorological variables adapted from the NWFP EC processing report [25]

Require: Primary EC series $x_{1:T}$; auxiliary series $z_{1:T}$; timestamps $t_{1:T}$; variable name v **Ensure:** Gap-filled series $\hat{x}_{1:T}$

```
1:  $\hat{x} \leftarrow x$ 
▷ Variable-specific quality control
2: if  $v = \text{PA}$  then
3:   Decompose  $x$  using STL with hourly frequency
4:   Detect anomalies in the decomposition remainder
5:   Replace detected anomalies in  $x$  with NaN
6: else if  $v = \text{LE}$  then
7:   Replace the manually identified outlier with NaN
8: else if  $v = \text{TA}$  then
9:   Convert the primary temperature to the units of the auxiliary series
10: end if
▷ Variables replaced directly by auxiliary measurements
11: if  $v = \text{SWC}$  then
12:    $\hat{x} \leftarrow z^{\text{COSMOS},1}$ 
13:   return  $\hat{x}_{1:T}$ 
14: else if  $v = \text{TS}$  then
15:    $\hat{x} \leftarrow z^{\text{COSMOS},2}$ 
16:   return  $\hat{x}_{1:T}$ 
17: end if
▷ Fit calibration model using paired observations
18:  $\mathcal{I} \leftarrow \{t : x_t \neq \text{NaN} \wedge z_t \neq \text{NaN}\}$ 
19: Fit
▷ Preserve measurements and fill missing values

$$x_t = \beta_0 + \beta_1 z_t + \varepsilon_t, \quad t \in \mathcal{I}$$

20: for  $t \leftarrow 1$  to  $T$  do
21:   if  $x_t \neq \text{NaN}$  then
22:      $\hat{x}_t \leftarrow x_t$ 
23:   else if  $z_t \neq \text{NaN}$  then
24:      $\hat{x}_t \leftarrow \hat{\beta}_0 + \hat{\beta}_1 z_t$ 
25:   else
26:      $\hat{x}_t \leftarrow \text{NaN}$ 
27:   end if
28: end for
29: return  $\hat{x}_{1:T}$ 
```

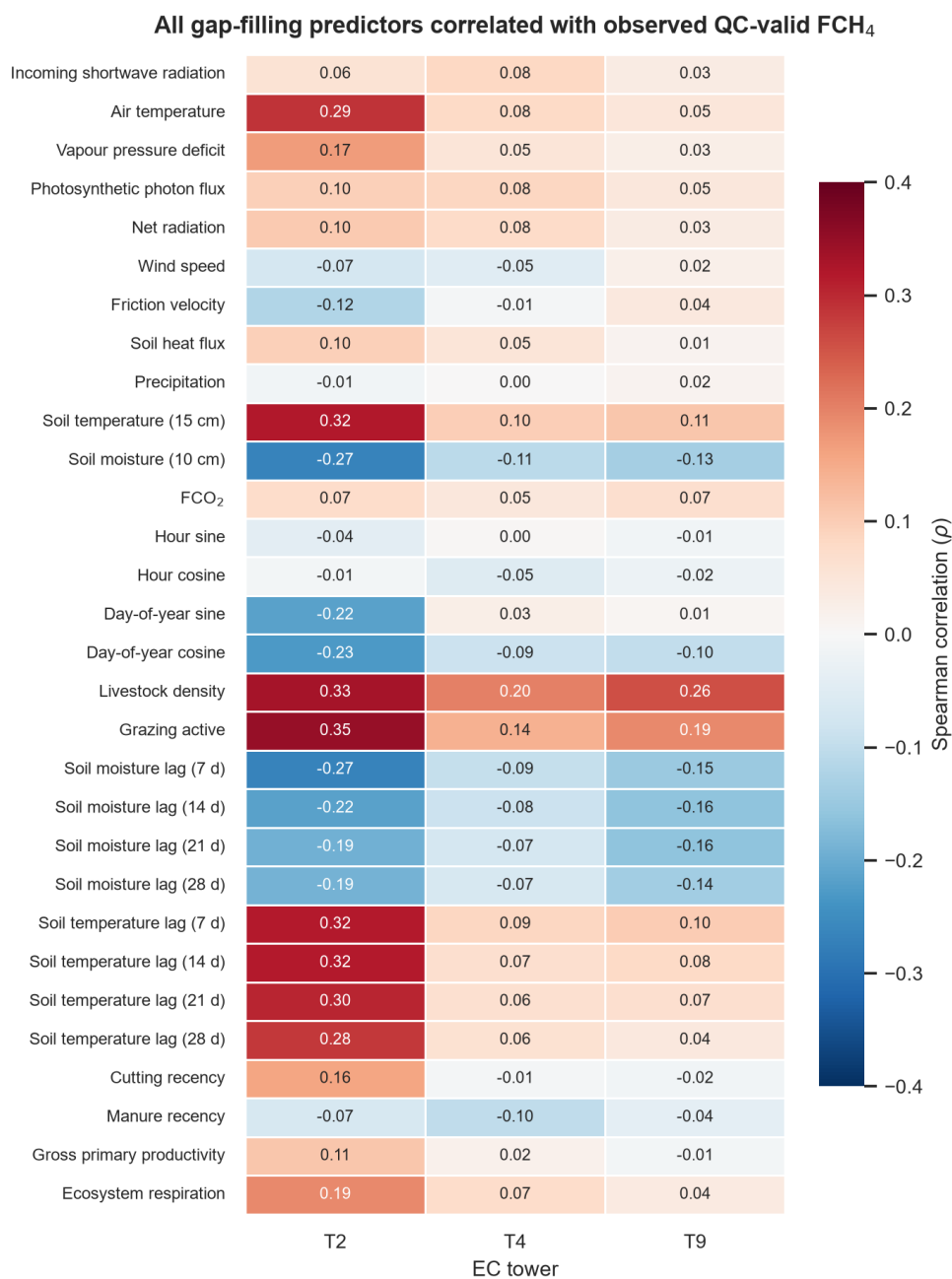


Figure A.1: Complete matrix of tower-specific Spearman correlations between observed QC-valid FCH₄ and all 30 predictors used by the gap-filling experiments. The reduced matrix in Figure 3.2 presents the union of the five largest absolute correlations within each tower. These coefficients describe pairwise monotonic association and should not be interpreted as causal effects or model-based feature importance.

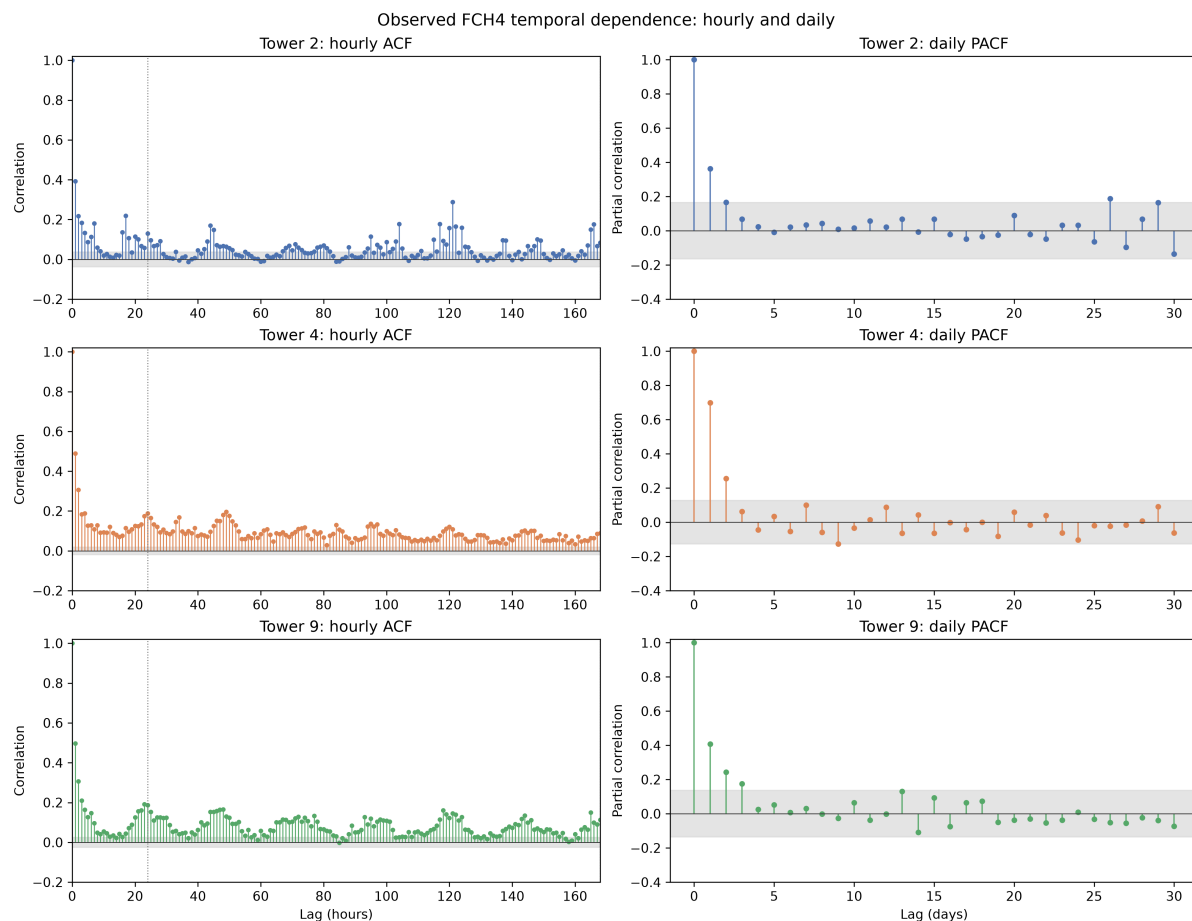


Figure A.2: Temporal dependence of QC-valid observed FCH₄. Left: hourly ACF calculated from available observation pairs at each lag, with 24 hours marked by the dotted line. Right: daily PACF calculated over the longest sufficiently complete segment at each tower; shaded regions show approximate 95% confidence bounds.

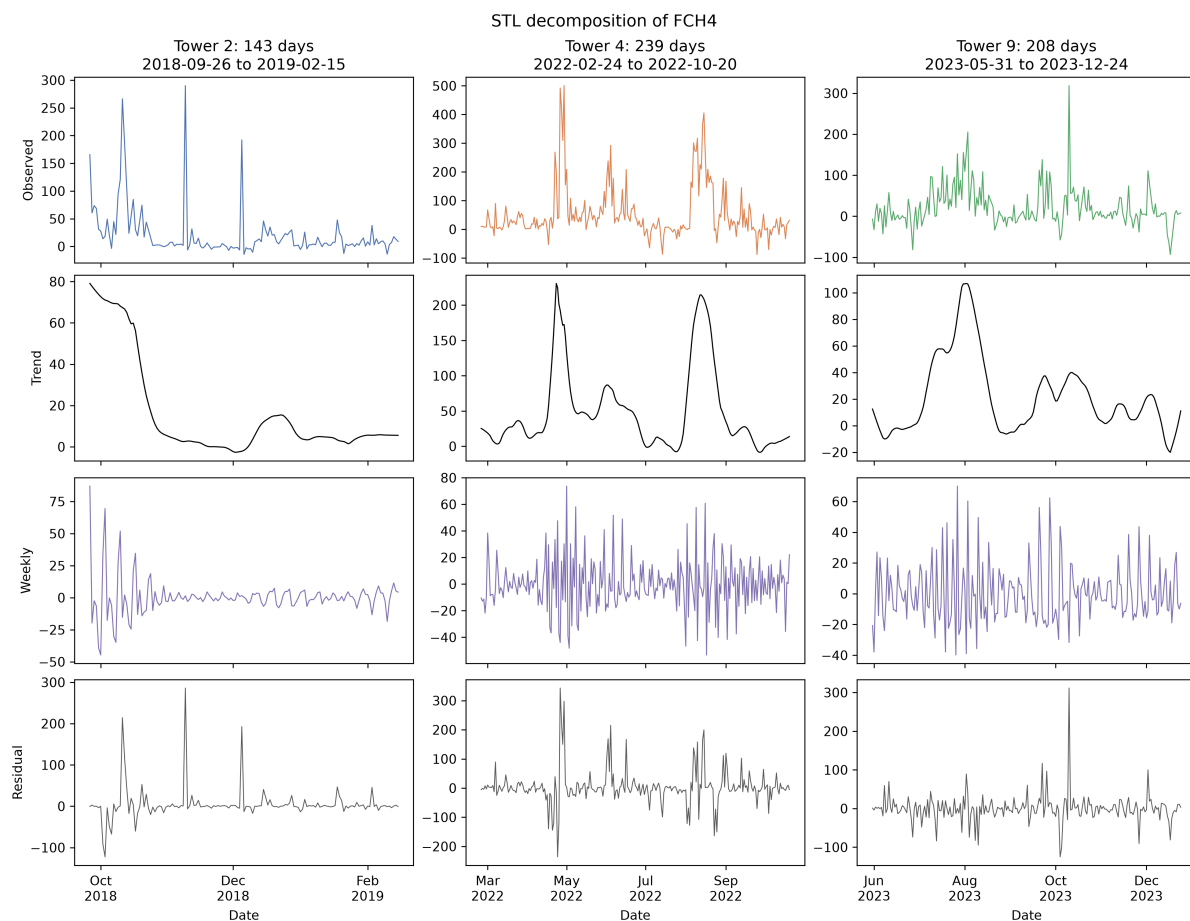


Figure A.3: Weekly STL decomposition of the longest sufficiently complete daily FCH₄ segment at each tower. Rows show the observed series, trend, weekly component and residual.

B | Models

This appendix supplements the operating-principle summary in Table 4.2 with model-specific pseudocode. The common synthetic-gap and fixed-origin forecasting protocols remain defined in Chapter 4; they are not repeated here. Likewise, Algorithm 1 already documents auxiliary environmental-driver reconstruction. The algorithms below therefore focus on how each benchmark model learns or produces predictions. “GF” denotes gap-filling and “FC” denotes forecasting.

B.1 Multivariate Imputation

MICE and HyperImpute treat FCH_4 as an incomplete column in a joint feature–target matrix. MICE uses a fixed Bayesian-ridge conditional model [31], whereas HyperImpute selects a conditional learner separately for each variable through internal validation [32].

Algorithm 2 MICE reconstruction with Bayesian-ridge conditional models

Require: Pooled incomplete matrix $Z = [y, X, \text{tower indicators}]$; held-out target positions H

Ensure: Reconstructed target $\hat{y}_H$

```

1: Set  $Z_{H,y}$  to missing before constructing the imputer input
2: Initialise each incomplete column from its observed marginal values
3: for chained iteration  $q \leftarrow 1$  to  $K$  do
4:   for each incomplete column  $j$  do
5:     Define observed rows  $O_j$  and missing rows  $M_j$ 
6:     Fit Bayesian ridge  $f_j^{(q)} : Z_{O_j,-j} \mapsto Z_{O_j,j}$ 
7:     Update  $Z_{M_j,j} \leftarrow f_j^{(q)}(Z_{M_j,-j})$ 
8:   end for
9:   if the iterative-imputer convergence rule is satisfied then
10:     Break
11:   end if
12: end for
13: Extract  $\hat{y}_H \leftarrow Z_{H,y}$  using the held-out rows’ absolute positions
14: return  $\hat{y}_H$ 

```

Algorithm 3 HyperImpute reconstruction with variable-specific AutoML models

Require: Pooled incomplete matrix $Z = [y, X, \text{tower indicators}]$; held-out positions H ; candidate learner library $\mathcal{L}$ **Ensure:** Reconstructed target $\hat{y}_H$

```

1: Set  $Z_{H,y}$  to missing and initialise incomplete columns
2: for each chained-imputation round do
3:   for each incomplete column  $j$  do
4:     Determine whether  $Z_j$  requires regression or classification
5:     for each compatible learner  $f \in \mathcal{L}$  do
6:       Estimate internal validation loss  $L_j(f)$  on observed rows  $O_j$ 
7:     end for
8:     Select  $f_j^* \leftarrow \arg \min_{f \in \mathcal{L}} L_j(f)$ 
9:     Refit  $f_j^*$  on  $O_j$  and update  $Z_{M_j,j} \leftarrow f_j^*(Z_{M_j,-j})$ 
10:   end for
11:   if the imputation sequence converges or reaches its iteration limit then
12:     Break
13:   end if
14: end for
15: return the completed target values  $\hat{y}_H$ 

```

B.2 Tree-Based Models

Random Forest, XGBoost and LightGBM were used in both tasks. For gap-filling, the query rows are the deliberately masked hours. For forecasting, each model is fitted at a forecast origin and its predicted FCH₄ value is appended to the available target history before constructing the next day’s autoregressive features. Preprocessing is always estimated from permitted training rows. The algorithms follow Breiman [33], Chen and Guestrin [34], and Ke et al. [35]. The unweighted voting-regression ensemble then averages the three aligned forecast chains following the forecast-combination principle of Bates and Granger [43].

Algorithm 4 Random Forest reconstruction or recursive forecasting

Require: Training rows (X_O, y_O) ; query rows X_Q ; number of trees B ; minimum leaf size m ; task $\tau \in \{\text{GF}, \text{FC}\}$

Ensure: Predictions $\hat{y}_Q$

- 1: Fit mean imputation on X_O and apply it to X_O and X_Q
 - 2: Use $(B, m) = (500, 5)$ for GF and $(500, 10)$ for FC
 - 3: **for** $b \leftarrow 1$ to B **do**
 - 4: Draw a bootstrap sample of (X_O, y_O)
 - 5: Grow T_b using squared-error reduction and random candidate features
 - 6: Stop a branch when further splitting violates the minimum leaf size
 - 7: **end for**
 - 8: Define $f(x) \leftarrow B^{-1} \sum_{b=1}^B T_b(x)$
 - 9: **if** $\tau = \text{GF}$ **then**
 - 10: Predict the masked hourly rows directly: $\hat{y}_Q \leftarrow f(X_Q)$
 - 11: **else**
 - 12: **for** forecast day $h \leftarrow 1$ to 365 **do**
 - 13: Build day- h features from known covariates and available target history
 - 14: Predict $\hat{y}_h \leftarrow f(x_h)$ and append $\hat{y}_h$ to that history
 - 15: **end for**
 - 16: **end if**
 - 17: **return** $\hat{y}_Q$
-

Algorithm 5 XGBoost reconstruction or recursive forecasting

Require: Training rows (X_O, y_O) ; query rows X_Q ; boosting rounds B ; learning rate η

Ensure: Predictions $\hat{y}_Q$

- 1: Fit training-only predictor imputation and initialise $F_0(x)$ with a constant
 - 2: **for** $b \leftarrow 1$ to B **do**
 - 3: **for** each $i \in O$ **do**
 - 4: Compute $g_i = \partial_F \ell(y_i, F_{b-1}(x_i))$ and $h_i = \partial_F^2 \ell(y_i, F_{b-1}(x_i))$
 - 5: **end for**
 - 6: Grow T_b by maximising regularised split gain from sums of g_i and h_i
 - 7: Assign each leaf its regularised optimum weight
 - 8: Update $F_b(x) \leftarrow F_{b-1}(x) + \eta T_b(x)$
 - 9: **end for**
 - 10: Use 500 rounds and minimum child weight 5 for GF
 - 11: Use 400 rounds, depth 2, $\eta = 0.02$ and minimum child weight 10 for FC
 - 12: Predict GF rows directly or generate the FC chain by appending each prediction
 - 13: **return** $F_B(X_Q)$ or the resulting recursive chain
-

Algorithm 6 LightGBM reconstruction or recursive forecasting

Require: Training rows (X_O, y_O) ; query rows X_Q ; boosting rounds B ; learning rate η **Ensure:** Predictions $\hat{y}_Q$

- 1: Fit training-only predictor imputation and bin continuous predictors
 - 2: Initialise $F_0(x)$ with a constant
 - 3: **for** $b \leftarrow 1$ to B **do**
 - 4: Calculate loss gradients from y_O and $F_{b-1}(X_O)$
 - 5: Accumulate gradient statistics by feature-histogram bin
 - 6: **while** a permissible positive-gain split exists **do**
 - 7: Split the leaf giving the greatest loss reduction
 - 8: **end while**
 - 9: Update $F_b(x) \leftarrow F_{b-1}(x) + \eta T_b(x)$
 - 10: **end for**
 - 11: Use 500 rounds and minimum child size 5 for GF
 - 12: Use 400 rounds, 7 leaves, $\eta = 0.02$ and minimum child size 10 for FC
 - 13: Predict GF rows directly or recursively update the FC target history
 - 14: **return** $F_B(X_Q)$ or the resulting recursive chain
-

Algorithm 7 Unweighted voting-regression forecast ensemble

Require: Aligned Random Forest, XGBoost and LightGBM forecast chains**Ensure:** Equal-weight ensemble forecast $\hat{y}^{ens}$

- 1: **for** each tower, origin and forecast day t **do**
 - 2: Verify that all component predictions refer to the same target row
 - 3: $\hat{y}_t^{ens} \leftarrow (\hat{y}_t^{RF} + \hat{y}_t^{XGB} + \hat{y}_t^{LGBM})/3$
 - 4: **end for**
 - 5: Evaluate the ensemble only at QC-valid observed target rows
 - 6: **return** $\hat{y}^{ens}$
-

B.3 Neural Models

The ANN is retained as a published-baseline replication [36]. SAITS and the bidirectional LSTM use observations on both sides of historical gaps, whereas the forecasting LSTM, TFT and DLinear operate causally beyond each forecast origin [38, 39, 40, 41, 22].

Algorithm 8 Feed-forward ANN gap reconstruction

Require: Standardised observed pairs (X_O, y_O) ; masked queries X_H ; network layers 1: L **Ensure:** Reconstructed target $\hat{y}_H$

- 1: Initialise weights $W^{(\ell)}$ and biases $b^{(\ell)}$
 - 2: **for** each training epoch **do**
 - 3: Set $a^{(0)} \leftarrow X_O$
 - 4: **for** $\ell \leftarrow 1$ to $L - 1$ **do**
 - 5: $a^{(\ell)} \leftarrow \phi(W^{(\ell)}a^{(\ell-1)} + b^{(\ell)})$
 - 6: **end for**
 - 7: $\hat{y} \leftarrow W^{(L)}a^{(L-1)} + b^{(L)}$
 - 8: Back-propagate prediction loss and update all trainable parameters
 - 9: **end for**
 - 10: Apply the fitted network to X_H and reverse target scaling
 - 11: **return** $\hat{y}_H$
-

Algorithm 9 SAITS diagonally masked attention reconstruction

Require: Per-tower sequence $Z = [X, y]$ with y_H masked; 336-hour windows; 24-hour stride**Ensure:** Reconstructed target $\hat{y}_H$

- 1: Remove the eight explicit soil-lag channels and standardise the masked sequence
 - 2: Create overlapping windows and a chronological training-validation split
 - 3: Initialise 3 layers, $d_{model} = 256$, 4 heads, $d_k = d_v = 64$, $d_{ff} = 512$, dropout 0.1
 - 4: **for** epoch $e \leftarrow 1$ to 100 **do**
 - 5: Compute diagonally masked attention $A = \text{softmax}(QK^\top / \sqrt{d_k} + D)V$, where $D_{tt} = -\infty$
 - 6: Obtain reconstructions $\tilde{Z}^{(1)}$ and $\tilde{Z}^{(2)}$
 - 7: Combine them as $\tilde{Z} = \beta \odot \tilde{Z}^{(1)} + (1 - \beta) \odot \tilde{Z}^{(2)}$
 - 8: Minimise spike-weighted absolute reconstruction error in batches of 32
 - 9: Retain the best validation state; stop after 10 non-improving epochs
 - 10: **end for**
 - 11: Average overlapping estimates, reverse scaling and **return** values at H
-

Algorithm 10 Bidirectional LSTM gap reconstruction

Require: Standardised 336-hour windows $[X, y]$; observation mask m ; held-out set H **Ensure:** Reconstructed target $\hat{y}_H$

- 1: Replace missing target inputs by zero and append mask m_t as an input channel
 - 2: Initialise a two-layer Bi-LSTM with 128 hidden units per direction and dropout 0.1
 - 3: **for** epoch $e \leftarrow 1$ to 50 **do**
 - 4: Randomly mask 20% of observed targets to form reconstruction mask $\tilde{m}$
 - 5: $\vec{h}_t \leftarrow \text{LSTM}(u_t, \vec{h}_{t-1})$
 - 6: $\overleftarrow{h}_t \leftarrow \text{LSTM}(u_t, \overleftarrow{h}_{t+1})$
 - 7: $\hat{y}_t \leftarrow W[\vec{h}_t; \overleftarrow{h}_t] + b$
 - 8: Minimise $\sum_t \tilde{m}_t(1 + |y_t|)|y_t - \hat{y}_t| / \sum_t \tilde{m}_t$ using Adam and learning rate 10^{-3}
 - 9: Retain the best validation state; stop after 8 non-improving epochs
 - 10: **end for**
 - 11: Average overlapping estimates, reverse scaling and **return** values at H
-

Algorithm 11 Causal LSTM forecasting

Require: Pre-origin daily target and drivers; 28-day lookback; hidden size 64**Ensure:** Continuous 365-day forecast

- 1: Form training windows ending no later than the forecast origin
 - 2: Standardise using pre-origin training statistics
 - 3: **for** each training window and time t **do**
 - 4: Compute input, forget and output gates i_t, f_t, o_t
 - 5: $\tilde{c}_t \leftarrow \tanh(W_c u_t + U_c h_{t-1} + b_c)$
 - 6: $c_t \leftarrow f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$
 - 7: $h_t \leftarrow o_t \odot \tanh(c_t)$ and map h_t to the next target
 - 8: Update parameters by back-propagation through time
 - 9: **end for**
 - 10: **for** forecast day $h \leftarrow 1$ to 365 **do**
 - 11: Predict from the latest 28 permitted days and known day- h drivers
 - 12: Append the prediction, rather than the later observation, to the history
 - 13: **end for**
 - 14: Reverse target scaling and **return** the forecast chain
-

Algorithm 12 Temporal Fusion Transformer forecasting

Require: Pre-origin daily sequences; known-future covariates; 28-day lookahead; $d_{model} = 32$; 4 heads; dropout 0.1

Ensure: Continuous 365-day forecast and optional quantiles

- 1: Encode static, observed-history and known-future variables separately
 - 2: Apply variable-selection networks to obtain time-varying feature weights
 - 3: Pass selected inputs through gated residual networks and an LSTM encoder-decoder
 - 4: Apply causal temporal attention over permitted positions
 - 5: Produce point or quantile outputs through the gated output head
 - 6: **for** forecast day $h \leftarrow 1$ to 365 **do**
 - 7: Supply the latest 28-day context and covariates known for day h
 - 8: Retain the next prediction and append it to the target history
 - 9: **end for**
 - 10: **return** the continuous chain and any requested quantiles
-

Algorithm 13 DLinear decomposition forecasting

Require: Pre-origin daily sequence $z_{t-27:t}$; moving-average kernel 25

Ensure: Continuous 365-day forecast

- 1: Estimate trend $\tau \leftarrow \text{MovingAverage}_{25}(z)$
 - 2: Calculate residual component $s \leftarrow z - \tau$
 - 3: Learn linear projections W_τ, b_τ and W_s, b_s from training windows
 - 4: Define $\hat{z} \leftarrow (W_\tau \tau + b_\tau) + (W_s s + b_s)$
 - 5: **for** forecast day $h \leftarrow 1$ to 365 **do**
 - 6: Predict from the latest 28-day sequence
 - 7: Append the prediction and advance the input window by one day
 - 8: **end for**
 - 9: Reverse target scaling and **return** the continuous chain
-

B.4 Statistical and Kernel Models

Support-vector regression is retained as a literature-aligned gap-filling replication [37]. SARIMAX provides the forecasting benchmark with an explicit stochastic time-series structure [42].

Algorithm 14 ϵ -support-vector regression gap reconstruction

Require: Standardised training pairs (X_O, y_O) ; queries X_H ; kernel K ; penalty C ; tube width ϵ **Ensure:** Reconstructed target $\hat{y}_H$

- 1: Solve $\min_{w, b, \xi, \xi^*} \frac{1}{2} \|w\|^2 + C \sum_i (\xi_i + \xi_i^*)$ subject to $|y_i - (w^\top \phi(x_i) + b)| \leq \epsilon + \xi_i$
 - 2: Retain rows with non-zero dual coefficients as support vectors
 - 3: Define $\hat{y}(x) \leftarrow \sum_{i \in O} (\alpha_i - \alpha_i^*) K(x_i, x) + b$
 - 4: Reverse target scaling and **return** $\hat{y}(X_H)$
-

Algorithm 15 AIC-selected SARIMAX forecasting

Require: Per-tower pre-origin target y_t ; exogenous vector x_t ; $p \in \{1, 2\}$, $d = 1$, $q \in \{0, 1\}$ **Ensure:** Continuous 365-day forecast

- 1: **for** each candidate order $(p, 1, q)$ **do**
 - 2: Fit $\Phi_p(B)(1 - B)y_t = c + \beta^\top x_t + \Theta_q(B)\varepsilon_t$ by maximum likelihood on pre-origin observations
 - 3: Calculate $\text{AIC} = 2k - 2 \log \hat{L}$
 - 4: **end for**
 - 5: Select the converged candidate with minimum AIC
 - 6: Forecast 365 steps using its dynamic state update and known future x_t
 - 7: Propagate the state without inserting later observed FCH_4
 - 8: **return** predictive means and state-space intervals
-

B.5 Pretrained Tabular Models

B.5.1 Pretraining and in-context adaptation

TabPFN and TabICLv2 are prior-data fitted networks (PFNs). Their model parameters are learned once, before application to this study, across a large distribution of synthetic tabular prediction tasks. For a synthetic task $\mathcal{T} \sim p_{\text{synth}}$, rows are divided into a labelled context $\mathcal{D}_C = (X_C, y_C)$ and an unlabelled query X_Q with held-out target y_Q . Pretraining minimises the query negative log-likelihood,

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{\mathcal{T} \sim p_{\text{synth}}} \left[- \sum_{j \in Q} \log q_{\theta}(y_j \mid x_j, \mathcal{D}_C) \right], \quad (\text{B.1})$$

so that one forward pass subsequently approximates the posterior predictive rule induced by the synthetic task prior [15, 16, 18, 19]. This is amortised inference: the expensive learning occurs across synthetic tasks during offline pretraining, rather than through gradient optimisation on each new dataset.

TabPFN v2 retains a representation for every table cell. Numerical values are combined with feature-specific random embeddings, grouped in pairs, and passed through 12 Transformer blocks (192-dimensional embeddings and six attention heads). Each block alternates attention across features within a row and across samples within a feature. Context targets are embedded while query targets remain masked, allowing queries to attend to labelled examples without target leakage. For regression, the output head assigns probability to target intervals through a full-support bar distribution and is trained by negative log-likelihood; means, medians or quantiles are then derived from this distribution [16].

Its synthetic prior samples table properties and structural causal models: random directed acyclic graphs are populated with nonlinear mechanisms, transformations, feature types, missingness and noise. Approximately 100 million such tasks expose the model to varied tabular relationships. This study used the released generic `tabpfn-v2-regressor.ckpt`, not a time-series-fine-tuned checkpoint.

TabICLv2 instead compresses features before dataset-level inference. Repeated circular feature groups are given target-aware embeddings for labelled rows; a column Transformer processes each feature across samples, and a row Transformer combines the features into a 512-dimensional representation. A 12-layer dataset Transformer then relates labelled and query rows. Its query-aware scalable softmax (QASSMax) counters attention dilution as the context grows, reducing the principal scaling from approximately $O(n^2m + nm^2)$ for cell-level TabPFN to $O(n^2 + nm^2)$ [18, 19].

TabICLv2 is pretrained on approximately 35 million filtered synthetic tasks generated by

random graphs containing neural, tree, Gaussian-process, linear, quadratic, product and discretisation mechanisms. Its three-stage curriculum increases contexts from 1,024 to at most 60,000 rows and 100 features. The regression model uses Muon optimisation and pinball loss over 999 conditional quantiles. Thus, unlike squared-error regressors, neither TabPFN nor TabICLv2 directly optimises the downstream R^2 , RMSE, MAE or MASE metrics.

Neither checkpoint was pretrained on North Wyke Farm Platform data, methane fluxes or the held-out forecast years. Here, `fit()` only preprocesses and registers the chronologically permitted labelled context; it does not update θ^* . Temporal information therefore enters through context selection and the engineered calendar, environmental and management features. The direct forecasting configuration used eight internal estimators (TabPFN seed 137; TabICLv2 seed 42) without checkpoint fine-tuning. Forecast combinations follow Bates and Granger’s variance-reduction principle [43].

Algorithm 16 Published offline pretraining scheme for TabPFN and TabICLv2

Require: Model family $M \in \{\text{TabPFN}, \text{TabICLv2}\}$; synthetic task prior p_{synth}^M ; model q_θ **Ensure:** Released pretrained checkpoint θ^*

- 1: **for** each offline pretraining step **do**
 - 2: Sample table dimensions, feature types and task difficulty
 - 3: **if** $M = \text{TabPFN}$ **then**
 - 4: Sample an SCM directed acyclic graph, node mechanisms and noise
 - 5: **else**
 - 6: Sample a random graph with mixed neural, tree, GP and algebraic mechanisms
 - 7: **end if**
 - 8: Generate a synthetic table $\mathcal{T} = (X, y)$ and apply model-specific filtering
 - 9: Split rows into labelled context $\mathcal{D}_C$ and query (X_Q, y_Q)
 - 10: Predict $q_\theta(y_Q \mid X_Q, \mathcal{D}_C)$ in one forward pass
 - 11: **if** $M = \text{TabPFN}$ **then**
 - 12: Compute the discretised regression negative log-likelihood and update with Adam
 - 13: **else**
 - 14: Sum pinball losses over 999 quantiles and update with Muon
 - 15: Advance the three-stage table-size curriculum
 - 16: **end if**
 - 17: **end for**
 - 18: Freeze the final parameters; no study data enter this procedure
 - 19: **return** θ^* for subsequent in-context inference
-

Algorithm 17 TabPFN inference for gap filling or direct daily forecasting

Require: Labelled context $\mathcal{D}_C = \{(x_i, y_i)\}$; query features X_Q ; pretrained parameters θ ; task τ **Ensure:** Predictive distribution and point predictions

- 1: **if** $\tau = \text{GF}$ **then**
 - 2: Construct an hourly tabular task after masking y_H and exclude H from context
 - 3: **else**
 - 4: Use pre-origin daily rows and the 52-driver plus 5 tower/time-indicator frame
 - 5: Optionally restrict context to the most recent 1,095 or 1,460 days
 - 6: **end if**
 - 7: Fit training-only missing-value preprocessing and configured target scaling
 - 8: Jointly encode labelled context and unlabelled query rows with the pretrained model
 - 9: Infer $p_\theta(y_Q \mid \mathcal{D}_C, X_Q)$ without gradient-based task training
 - 10: Extract the configured point statistic (median for direct forecasts) and quantiles
 - 11: Reverse target scaling and **return** predictions
-

Algorithm 18 TabICLv2 inference for gap filling or direct daily forecasting

Require: Labelled context $\mathcal{D}_C$; query features X_Q ; pretrained parameters θ ; task τ **Ensure:** Point predictions and optional quantiles

- 1: **if** $\tau = \text{GF}$ **then**
 - 2: Train separately by tower on the expanded 30-predictor frame
 - 3: Exclude H and sample at most 10,000 observed rows using seed 42
 - 4: Mean-impute predictors using the sampled context
 - 5: **else**
 - 6: Use pre-origin daily context and the applicable forecasting frame
 - 7: Optionally retain all history, 1,095 days or 1,460 days
 - 8: Apply tower-wise target scaling for the configured 1,460-day variant
 - 9: **end if**
 - 10: Register the labelled context through the model fit interface
 - 11: Keep the pretrained checkpoint fixed; `fit()` performs no gradient update
 - 12: Infer $p_\theta(y_Q \mid \mathcal{D}_C, X_Q)$ jointly from context and queries
 - 13: Extract the median point prediction and optional 0.05, 0.50 and 0.95 quantiles
 - 14: Reverse target scaling and **return** predictions
-

Algorithm 19 Probability-gated spike correction for a foundation-model forecast

Require: Base regressor f_b ; spike classifier c ; spike regressor f_s ; normal regressor f_n ; pre-origin tower-specific 95th percentile $q_{0.95}$ **Ensure:** Corrected forecast $\hat{y}^{corr}$

- 1: Label context row i as a spike if $y_i > q_{0.95}$
 - 2: Fit or condition c , f_s , f_n and f_b using permitted context only
 - 3: **for** each query day t **do**
 - 4: $p_t^{spike} \leftarrow c(x_t)$
 - 5: $e_t \leftarrow [f_s(x_t) - f_n(x_t)]_+$
 - 6: $\hat{y}_t^{corr} \leftarrow f_b(x_t) + 0.25 p_t^{spike} e_t$
 - 7: **end for**
 - 8: **return** $\hat{y}^{corr}$
-

Algorithm 20 Best-performing TabICL forecasting model

Require: TabICLv2 forecaster F ; pre-origin daily data; 365-day queries

Ensure: Final blended forecast $\hat{y}^{\text{best}}$

- 1: $\hat{y}^{(1)} \leftarrow$ all-history forecast from F with Algorithm 19
 - 2: $\hat{y}^{(2)} \leftarrow$ forecast from F using the latest 1,095 context days
 - 3: $\hat{y}^{(3)} \leftarrow$ forecast from F using the latest 1,460 context days and tower-wise target scaling
 - 4: **for** each forecast day t **do**
 - 5: $\hat{y}_t^{\text{best}} \leftarrow (\hat{y}_t^{(1)} + \hat{y}_t^{(2)} + \hat{y}_t^{(3)})/3$
 - 6: **end for**
 - 7: Retain the three component chains and evaluate the blend on identical observed rows
 - 8: **return** $\hat{y}^{\text{best}}$
-

C | Gap-Filling

Table C.1 records the source-specific methane gap-filling adaptations removed from the main chapter. Differences in ecosystem, predictors, measurement period and validation design mean that the reported source values provide context rather than controlled performance comparisons [8, 14, 9].

Replication	Model	NWFP R^2	Cited R^2
Irvin et al. [†] (17-site wetland benchmark)	RF	0.122	0.79
	XGBoost	0.125	$\approx 0.65\text{--}0.67$
Zhu et al. (UK pasture)[‡]	MDS	0.055	$\approx 0.03\text{--}0.05$
	RFm	0.054	< 0.10
Kim et al. [¶] (wetland/rice paddy)	MDS	0.032	$\approx 0.06\text{--}0.71$
	RF	0.135	$\approx 0.82\text{--}0.97$
	RF + PCA7	0.103	$\approx 0.76\text{--}0.96$
	ANN	0.139	$\approx 0.64\text{--}0.94$
	SVM	0.057	$\approx 0.33\text{--}0.82$

Table C.1: Contextual adaptations of published methane gap-filling baselines. NWFP values are averaged equally across the towers included by each adaptation; cited values are those reported by the corresponding source. Only Zhu et al. is comparable to the NWFP pasture ecosystem.

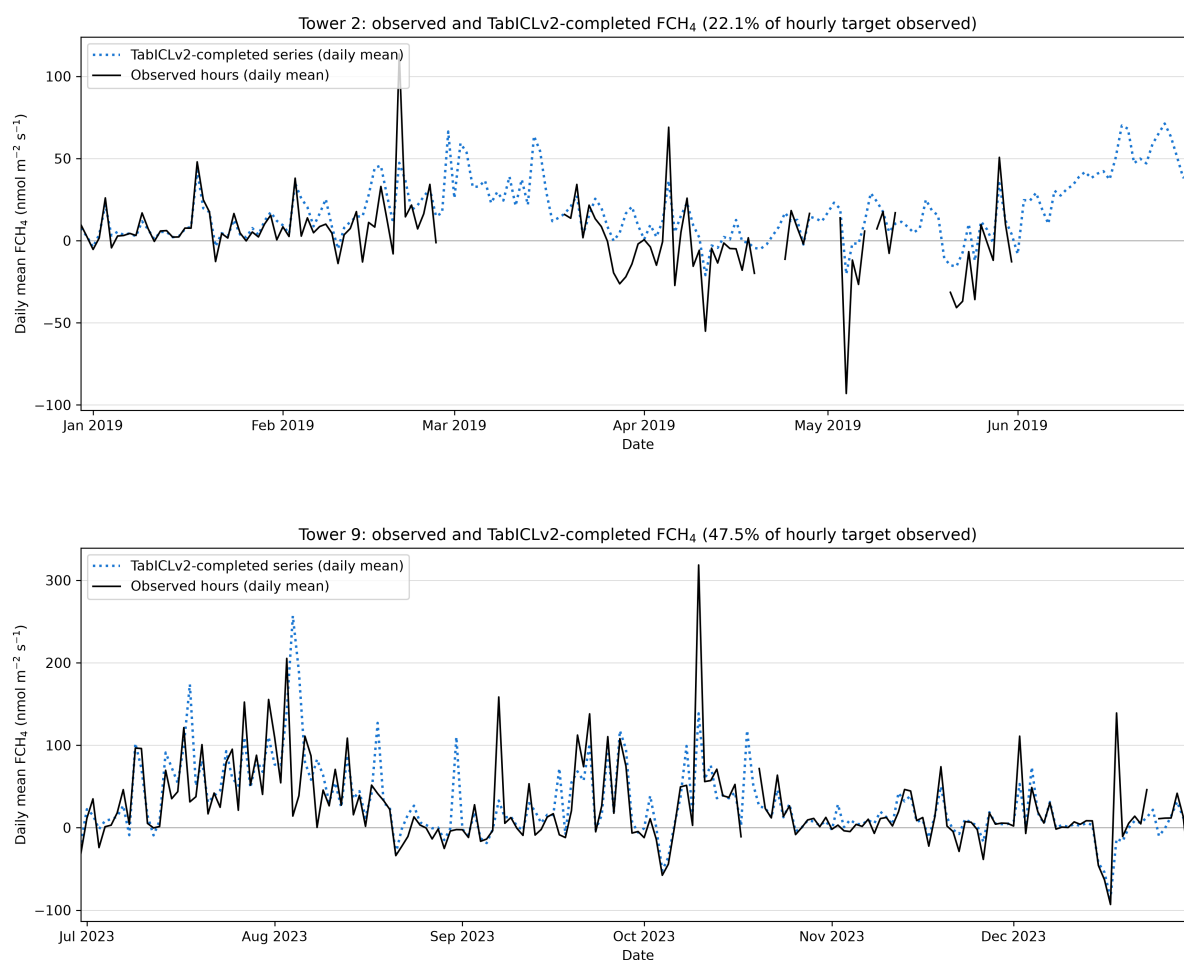


Figure C.1: Daily means from QC-valid observed hours (solid black) and the TabICLv2-completed series (dotted blue) at T2 during January–June 2019 (top) and T9 during July–December 2023 (bottom). Hourly target availability was 22.1% at T2 and 47.5% at T9. The completed series preserves observations and fills missing hours with TabICLv2 estimates; the panels are reconstruction diagnostics rather than held-out validation. The corresponding T4 example is shown in Figure 5.1.

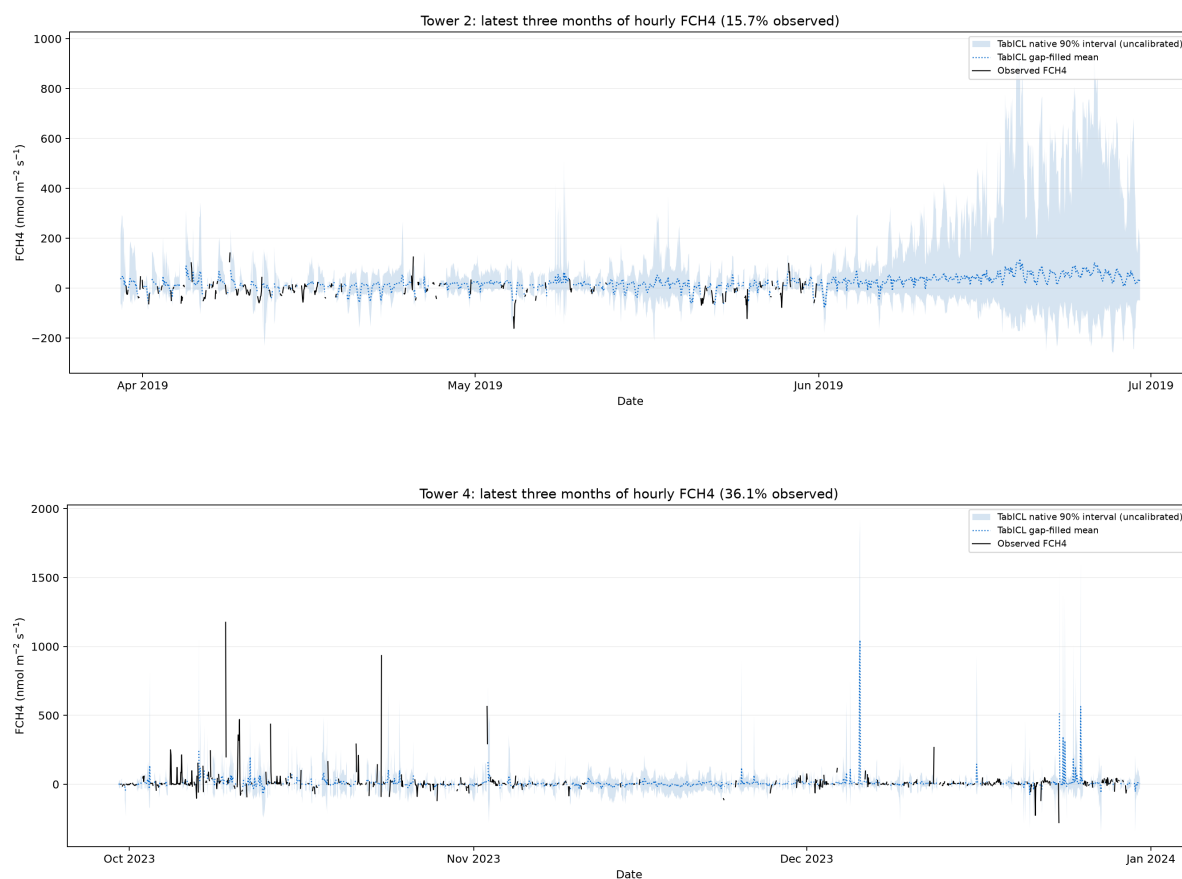


Figure C.2: Three-month examples of hourly FCH₄ reconstruction and native uncertainty for T2 (top) and T4 (bottom). Black lines show QC-valid observations, dotted blue lines show the TabiCLv2 gap-filled mean, and shaded regions show the uncalibrated native 90% interval. Each tower uses its latest available three-month window and its own vertical scale.

Model or experiment	RMSE	MAE
Training-set mean	134.39	53.01
MDS	140.58	53.67
RF (meteorology features only)	129.47	51.04
MICE	125.88	59.08
HyperImpute	105.90	45.85
RF (additional features)	94.14	41.11
LightGBM	97.55	43.57
XGBoost	103.17	49.21
SAITS	105.35	44.44
Bidirectional LSTM	112.41	45.67
TabICLv2	95.60	40.69

Table C.2: Aggregate error metrics for the gap-filling models in Table 5.2. Values are equal-weight averages of the three tower-specific median scores across the common synthetic-gap settings. RMSE and MAE are reported in $\text{nmol m}^{-2} \text{s}^{-1}$; lower values are preferable.

D | Forecasting

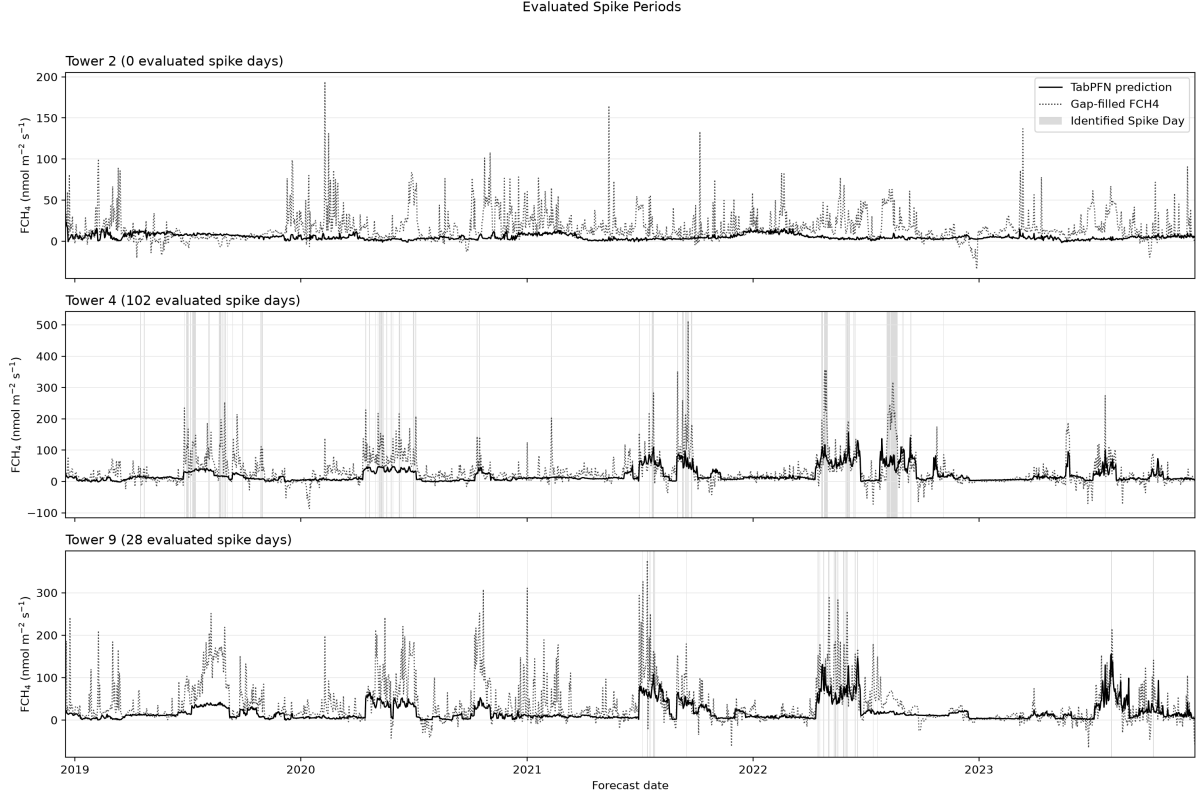


Figure D.1: Temporal distribution of the spike observations used in the forecasting evaluation. Grey bands mark QC-valid observations above the tower- and origin-specific pre-forecast 95th-percentile threshold; the solid line shows the TabPFN prediction and the dotted line shows the gap-filled FCH₄ series for context. The evaluated sample contains no spike days at T2, 102 at T4 and 28 at T9.

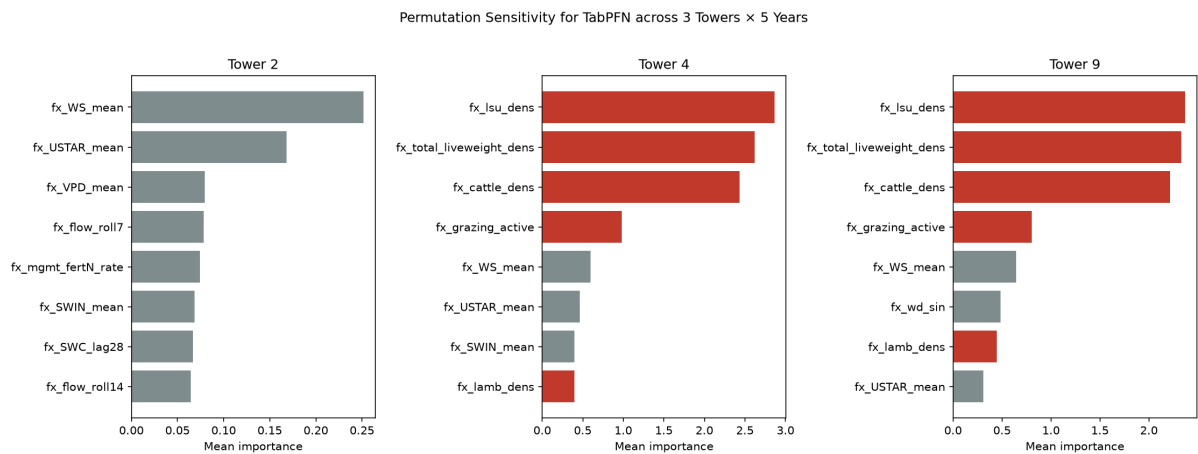


Figure D.2: Tower-specific permutation-sensitivity rankings for the TabPFN model. The figure shows the leading predictors within T2, T4 and T9, complementing the pooled ranking in Figure 6.2.

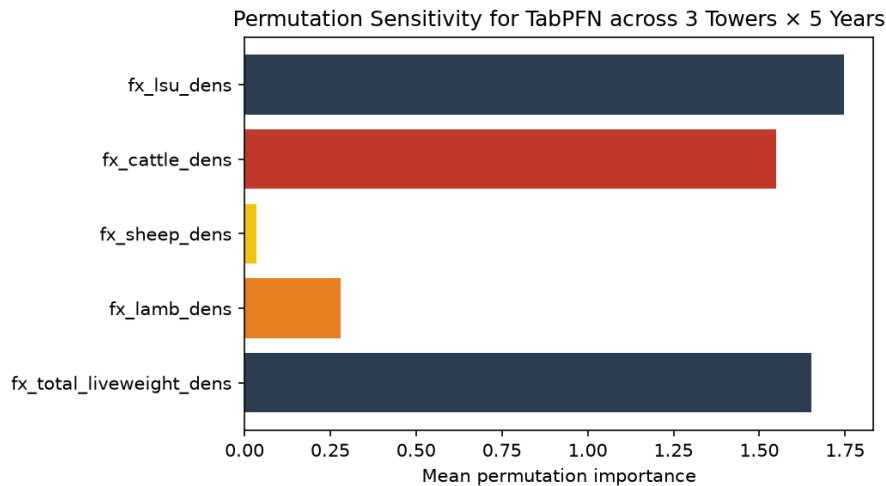


Figure D.3: Permutation sensitivity of the livestock-species predictors in the TabPFN model. The comparison separates the contributions of cattle, sheep and lamb density within the broader livestock signal.

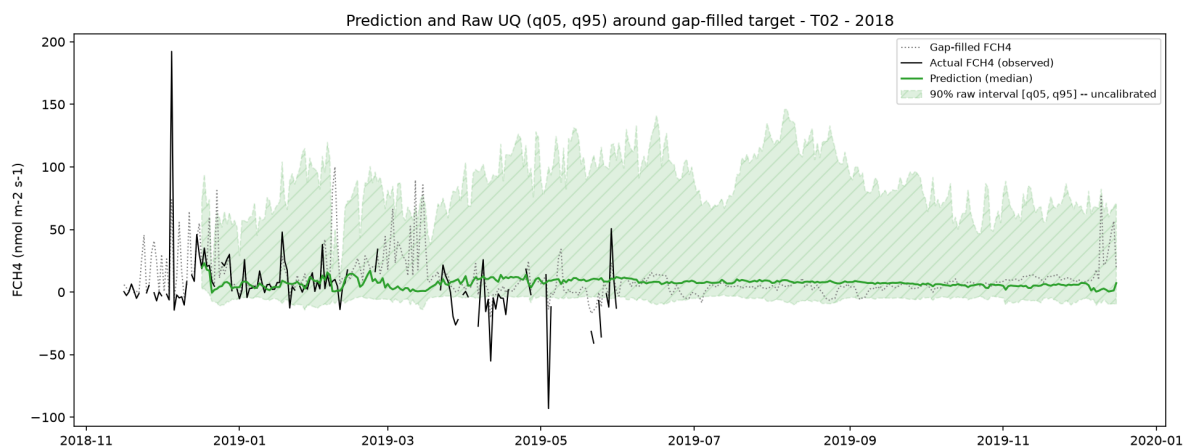


Figure D.4: Representative 365-day TabPFN forecast at T2 from the 2018 forecast origin. The green line is the median forecast, the black line shows available observations, and the dotted line provides the gap-filled series as context. The hatched region is the native raw 90% quantile interval. Conformal calibration was not applied because the available T2 observations were insufficient to construct a leave-one-origin-out calibration set.

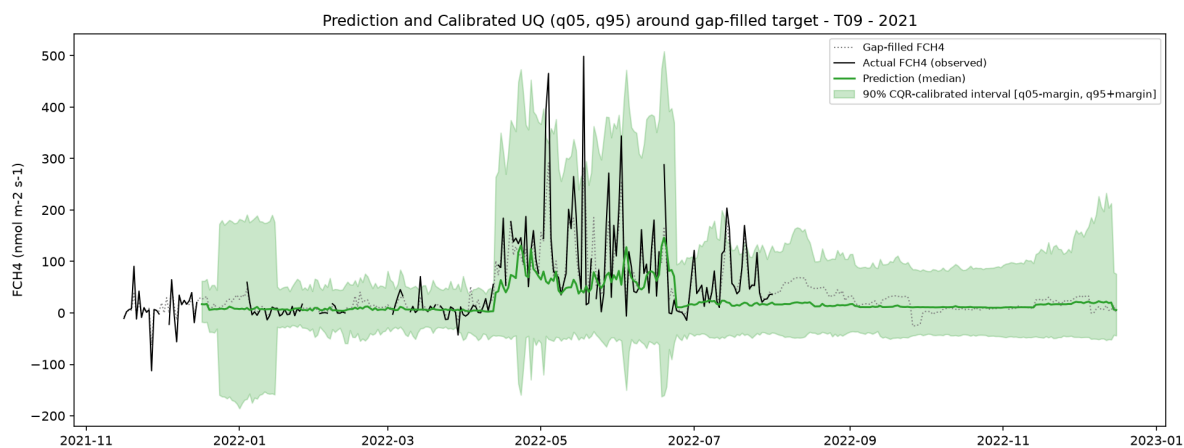


Figure D.5: Representative 365-day TabPFN forecast at T9 from the 2021 forecast origin. The green line is the median forecast, the black line shows available observations, the dotted line provides the gap-filled series as context, and the shaded region is the 90% CQR-calibrated prediction interval across the complete forecast horizon.

E | Scenario Projection

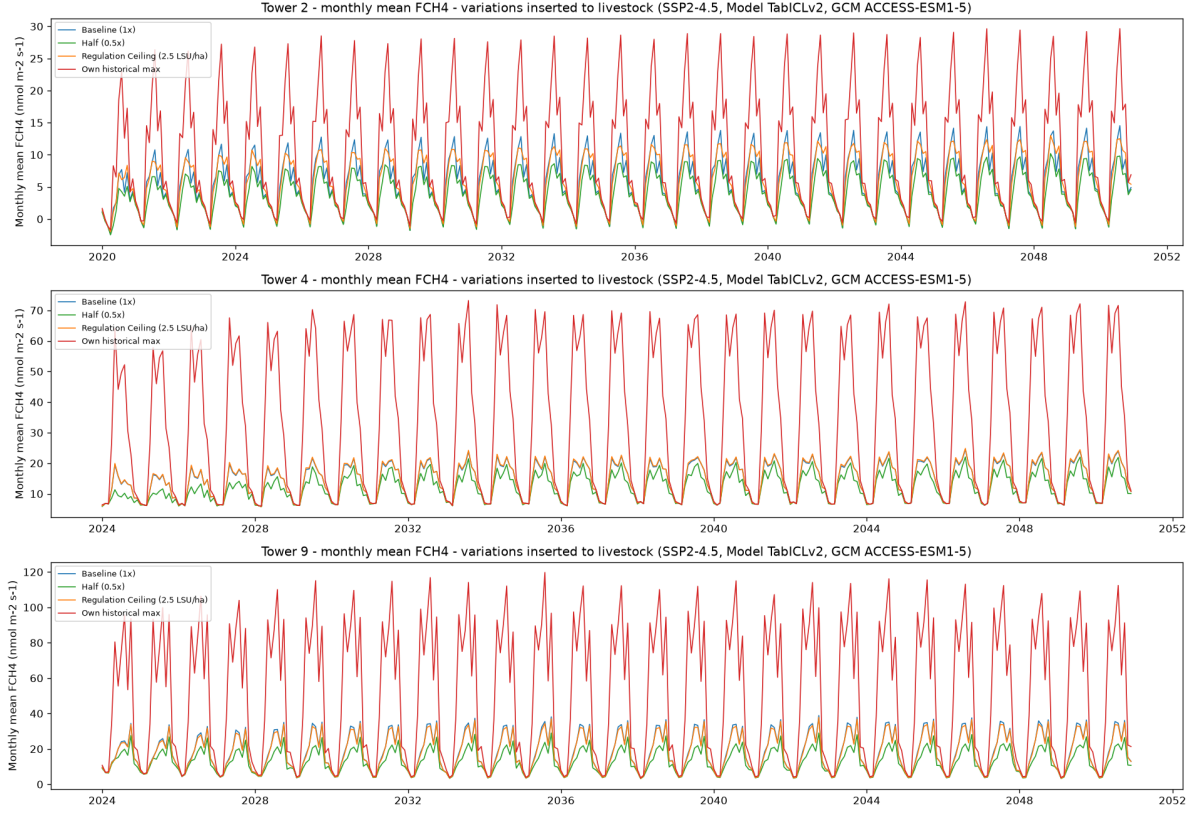


Figure E.1: Monthly mean FCH₄ (nmol m⁻² s⁻¹) at T2, T4 and T9 under SSP2-4.5 for the livestock-density ladder – baseline, half-density, government-reference and own-historical-maximum (all-species) – for a single illustrative trajectory (restricted model, GCM ACCESS-ESM1-5, realisation 1). Table 7.3 reports the mean annual percentage change, averaged across all five GCMs and ten realisations per GCM.

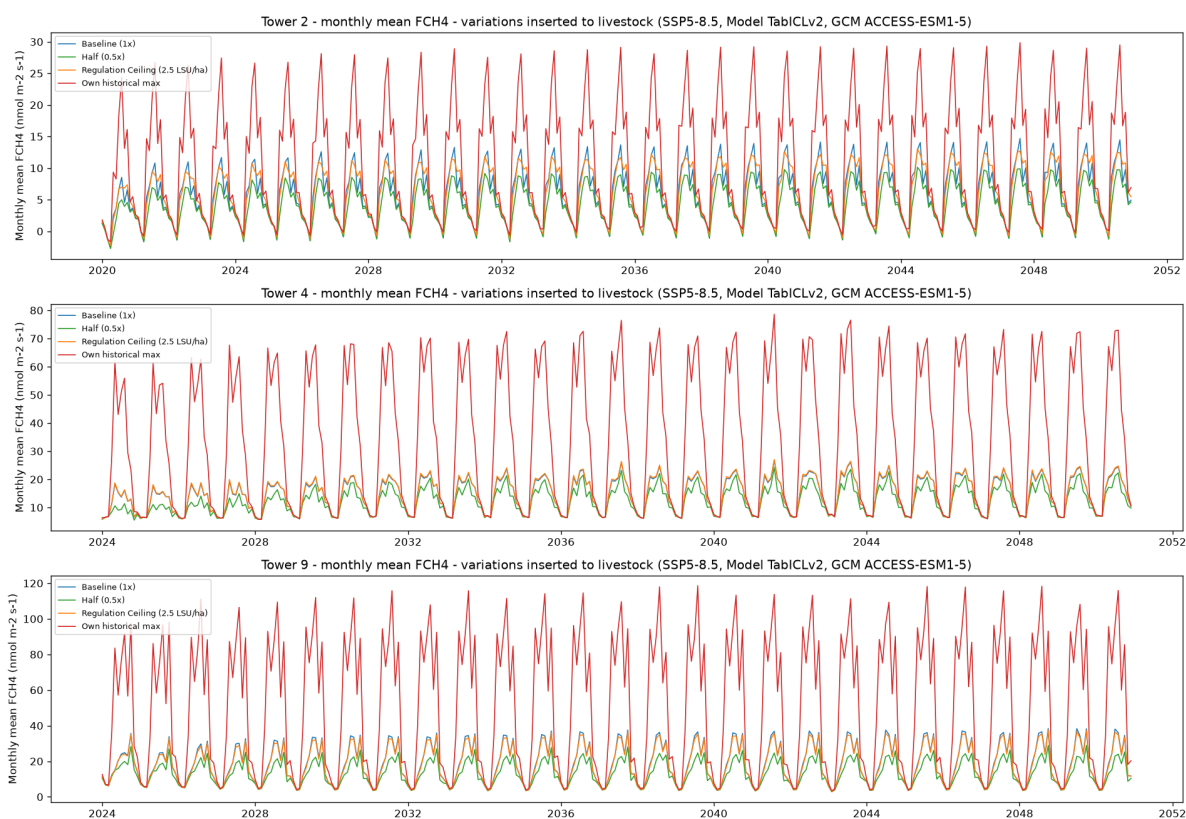


Figure E.2: As Figure E.1, under SSP5-8.5. The near-identical trajectories under both pathways illustrate Table 7.3's finding that the SSP pathway itself makes almost no difference to the livestock-density response.

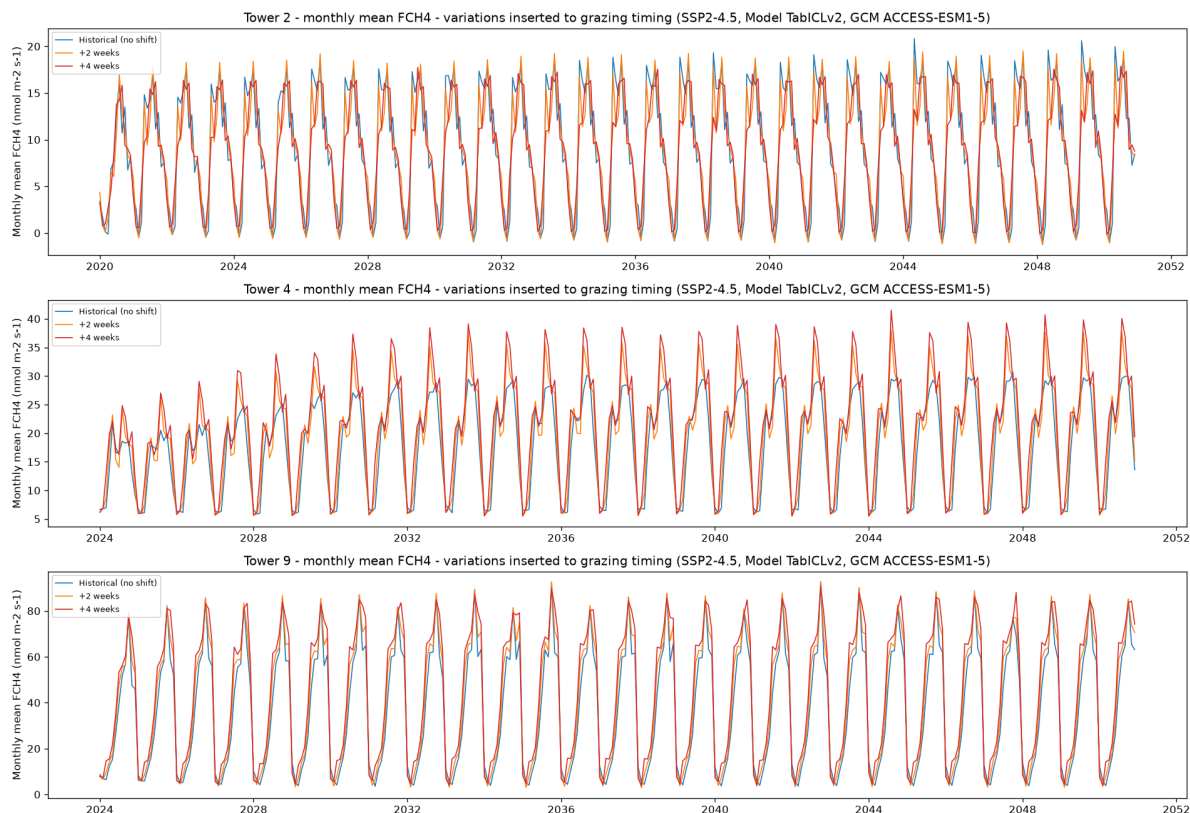


Figure E.3: Monthly mean FCH₄ (nmol m⁻² s⁻¹) at T2, T4 and T9 under SSP2-4.5 for the historical grazing-timing baseline and its +2-week and +4-week shift perturbations, for a single illustrative trajectory (restricted model, GCM ACCESS-ESM1-5, realisation 1). Table 7.4 reports the mean annual percentage change for the +4-week shift only, averaged across all five GCMs and ten realisations per GCM.

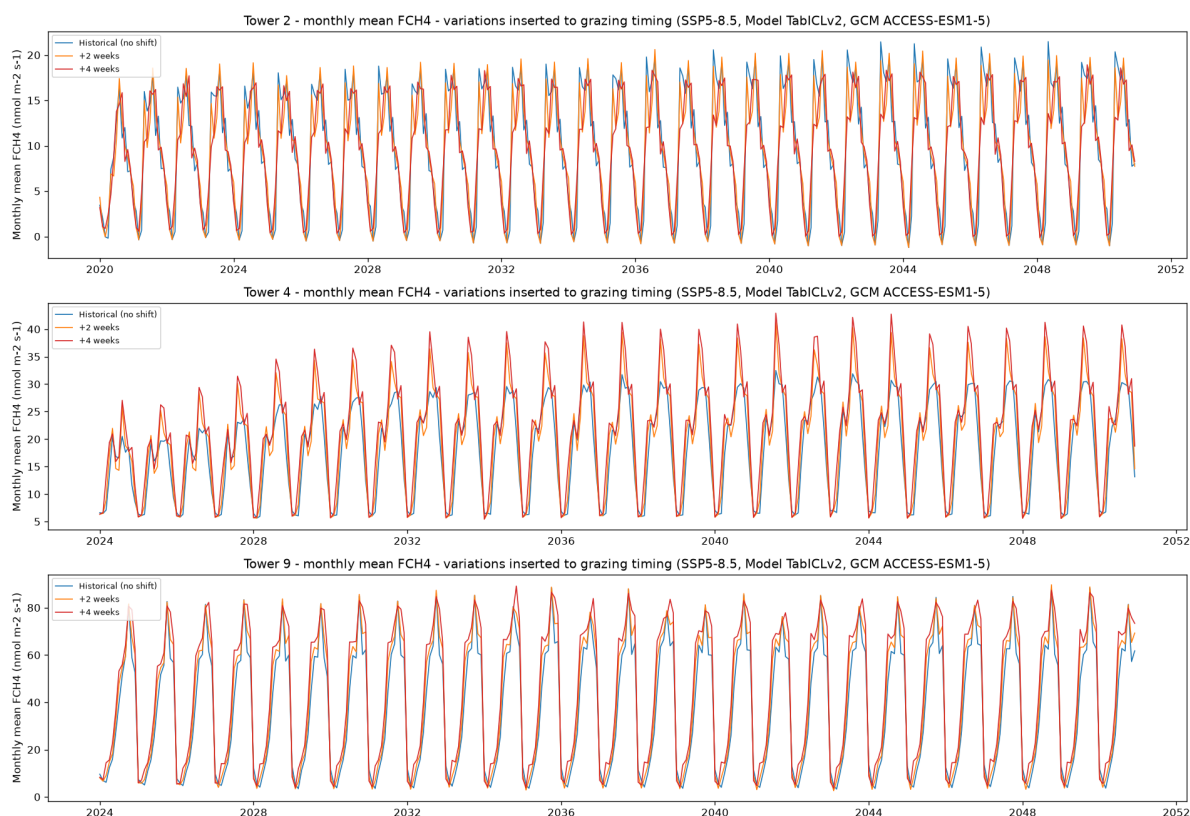


Figure E.4: Monthly mean FCH₄ (nmol m⁻² s⁻¹) at T2, T4 and T9 under SSP5-8.5 for the historical grazing-timing baseline and its +2-week and +4-week shift perturbations, for a single illustrative trajectory (restricted model, GCM ACCESS-ESM1-5, realisation 1). Table 7.4 reports the mean annual percentage change for the +4-week shift only, averaged across all five GCMs and ten realisations per GCM.

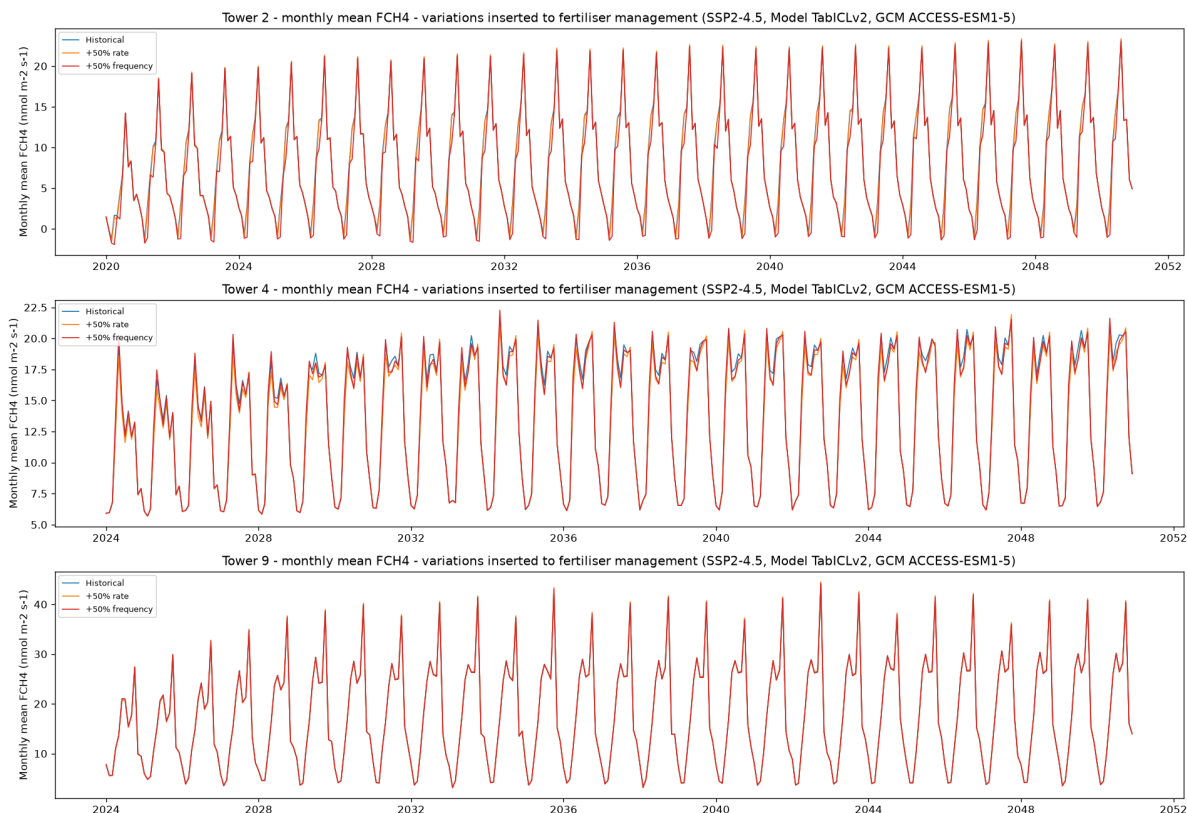


Figure E.5: Monthly mean FCH₄ (nmol m⁻² s⁻¹) at T2, T4 and T9 under SSP2-4.5 for the historical fertiliser-management baseline and its +50%-rate and +50%-frequency perturbations, for a single illustrative trajectory (restricted model, GCM ACCESS-ESM1-5, realisation 1). The three lines are visually near-identical at every tower, consistent with Table 7.4's finding that fertiliser interventions produce only a small effect on predicted FCH₄.

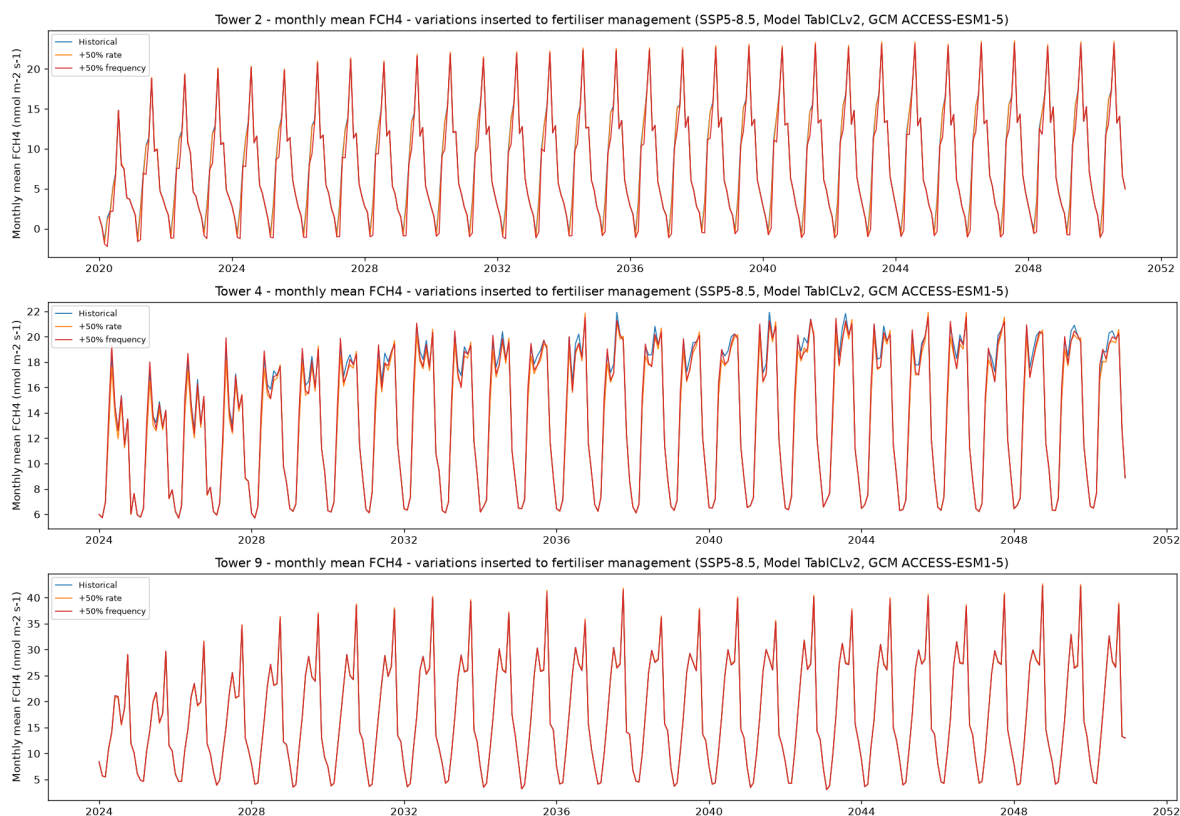


Figure E.6: Monthly mean FCH₄ (nmol m⁻² s⁻¹) at T2, T4 and T9 under SSP5-8.5 for the historical fertiliser-management baseline and its +50%-rate and +50%-frequency perturbations, for a single illustrative trajectory (restricted model, GCM ACCESS-ESM1-5, realisation 1). The three lines are visually near-identical at every tower, consistent with Table 7.4's finding that fertiliser interventions produce only a small effect on predicted FCH₄.

F | Source Code

Source code for all of the methods implemented in Chapters 3–7 can be found in the GitHub repository:

<https://github.com/nicholas-tobias-00/COMP0191-Dissertation>.